

Article

Fatigue Life Evaluation of a Steel–Prestressed Concrete Hybrid Tower Under Prestress Relaxation Using a Bidirectionally Coupled Damage Model and Neural Network Surrogate

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Abstract

This study investigates the long-term fatigue performance of an in-service 4.55 MW steel–prestressed concrete hybrid wind turbine tower under different tendon relaxation scenarios. The fatigue lives of prestressing tendons and concrete are calculated from finite element stress responses obtained in ABAQUS. A bidirectionally coupled damage model then updates the damage and stiffness of the two materials to determine the overall coupled fatigue life. Based on these results, a Spatial-Topology and Physics-Informed Cascade Network (STPI-Net) is developed for life prediction. The results show that fatigue life reductions vary across tendon relaxation cases and generally increase with prestress loss. Windward-side and simultaneous multi-tendon relaxation cause larger reductions, and damage coupling further reduces overall fatigue life in all studied cases. At 80% prestress loss, the overall coupled fatigue life reaches 0.45 years in the severe three-tendon relaxation case, representing a 28.6% reduction relative to uncoupled life. Scenario-based five-fold cross-validation is further used to evaluate the predictive performance of STPI-Net. The R^2 values are 0.8879 for logarithmic tendon fatigue life, 0.9030 for logarithmic concrete fatigue life, and 0.8195 for overall coupled fatigue life. The proposed framework supports the assessment of component and overall fatigue performance under different prestress relaxation conditions.

Keywords: steel–concrete hybrid tower; prestress relaxation; fatigue life evaluation; bidirectional damage coupling; physics-informed neural network

1. Introduction

As wind energy becomes more important to the global renewable energy transition, installed wind power capacity keeps expanding rapidly. At the same time, wind turbines are being developed with higher rated capacities and greater hub heights. Steel–prestressed concrete hybrid towers are increasingly used for large wind turbines, since they combine the construction efficiency of steel structures with the high stiffness and crack resistance of prestressed concrete. For this type of hybrid tower, prestressing tendons provide the concrete portion with pre-compression. This precompression improves the composite action between tendons and concrete, and also increases overall stiffness and durability. However, during long-term service, the tendons may experience prestress loss caused by anchorage relaxation, thermal fluctuations, and extreme loading events. Prestress loss changes the internal stress state and fatigue response of the structure. Therefore, it is important to accurately assess the long-term fatigue performance of hybrid wind turbine towers under prestress loss to ensure the service safety of large wind power structures.



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In recent years, substantial research has been conducted on the structural performance and fatigue durability of wind turbine support structures. Alonso-Martinez et al. [1] and Seidel et al. [2] examined the structural behavior of bolted flange connections in wind turbine towers, demonstrating the necessity of considering nonlinear effects and geometric imperfections in connection analysis. Zhang et al. [3] investigated the mechanical performance of curved steel–concrete–steel composite walls for wind turbine towers, highlighting the role of bidirectionally coupled damage in structural behavior.

The fatigue performance of wind turbine towers is also closely related to the evolution of material damage. In concrete, performance degradation is usually related to stiffness degradation, crack propagation, and damage accumulation. Riyar et al. [4] and Wei et al. [5] review concrete fatigue behavior and related fatigue analysis methods. Their work covered fatigue crack growth, fracture processes, size effects, and life prediction methods based on S–N curves, fracture mechanics, and damage mechanics. Qin et al. [6] extended the microplane model to steel-fibre-reinforced concrete and linked fatigue history variables to stiffness degradation, indicating that stiffness loss can characterize progressive concrete fatigue damage. Saqif and El-Tawil [7] developed an elastoplastic damage constitutive model for UHPC using cyclic compression test data, revealing the progressive stiffness and strength degradation of concrete under repeated loading. Other studies have focused on the evaluation of concrete degradation. Hassan et al. [8] combined non-destructive testing measurements with a mix-specific regression model to predict concrete compressive strength and emphasized the importance of material-specific calibration. Imoh et al. [9] investigated concrete containing waste glass cullet and snail shell powder and showed that changes in material composition significantly affected its workability and mechanical strength. Studies on geomaterials have also provided insights into internal structural evolution and fracture development linked to macroscopic mechanical degradation [10,11].

Fatigue evolution is further affected by load characteristics, damage accumulation, and structural degradation. Therefore, researchers have improved and extended the traditional Palmgren–Miner linear cumulative damage rule. Sadeghi et al. [12] proposed a rainflow residual reconstruction method to compensate for lost low-frequency large-amplitude fatigue cycles caused by 10 min signal truncation for offshore wind supports. Haselbozchaloe et al. [13] indicated that nonlinear damage models consider load-order effects and critical-plane schemes handle non-proportional loads. Zhu et al. [14] developed a Markov-subset simulation fatigue reliability model, showing that long-term operation without inspection greatly raises failure probability. In addition to fatigue damage caused by loads, the effects of environmental degradation have also received increasing attention. Studies have shown that corrosion degradation and welding residual stress can accelerate fatigue deterioration by affecting structural conditions and crack propagation [15,16].

The high computational cost of fatigue analysis has also encouraged the development of data-driven surrogate models for the efficient prediction and assessment of wind turbine structures. Bai et al. [17] developed a residual neural network using hub-height wind speed statistics to reduce the need for repeated numerical simulations. Similarly, Santos et al. [18] incorporated fatigue-related physical information into a physics-guided neural network, achieving errors below 3% for long-term accumulated damage-equivalent moment estimation. Their results demonstrated the value of physical information in data-driven fatigue prediction. Some other studies have combined continuous-time Bayesian networks and Bayesian optimization to assess wind turbine blade fatigue reliability and predict material fatigue life [19,20]. These methods integrate finite element simulation, experimental data, and parameter uncertainty, improving prediction reliability. Data-driven methods also enhance real-time operations and maintenance (O&M). Studies [21–23] leverage SCADA data alongside Gaussian process regression, health index modeling, and neural

networks for turbine condition assessment and anomaly detection. For physical damage inspection, Li et al. [24] combined UAV imagery with an attention-enhanced grid CNN for autonomous concrete tower crack inspection. In addition, recent work scales these methodologies from single components to entire wind farms. Pacheco et al. [25] used two years of strain-monitoring data to develop a probabilistic farm-wide fatigue estimation framework, showing that monitoring information from selected turbines can support fatigue assessment across a wind farm.

Although existing studies have made considerable progress in fatigue damage modeling and data-driven life prediction for wind turbine structures, several issues remain unresolved. In particular, the prestress degradation effect of steel–prestressed concrete hybrid wind turbine towers has not been systematically investigated. The fatigue response of these towers is affected by the interaction among tendon relaxation, concrete damage evolution, and external loading. Stress redistribution caused by prestress loss may further change local fatigue development and overall fatigue life. However, the effects of different relaxation locations, relaxation modes, and prestress loss levels on the fatigue performance of hybrid towers remain unclear. The relationship between local component degradation and overall fatigue life also requires further investigation. In addition, an effective method for incorporating prestress loss information into fatigue life prediction has not yet been fully established for the condition assessment of in-service wind turbine towers.

To address the above issues, this study investigates the fatigue performance evolution of an in-service 4.55 MW steel–prestressed concrete hybrid wind turbine tower under different tendon relaxation cases. An integrated analysis framework is established that combines high-fidelity finite element simulation, fatigue damage assessment, bidirectional damage coupling analysis, and data-driven prediction. First, a refined three-dimensional finite element model of the hybrid tower is established in ABAQUS. Different tendon relaxation cases are then constructed to examine the effects of relaxation location and prestress loss level on fatigue performance. This paper mainly displays the cases of single-tendon relaxation (SR1–SR5) and multiple-tendon relaxation (MR1–MR4) under different prestress loss levels (20%, 40%, 60%, and 80%). These cases can reveal the influence of typical spatial degradation modes on structural performance. Based on the fatigue analysis results, a bidirectionally coupled damage model is established to describe the interaction between prestressing tendon and concrete. This model is used to describe the effects of stiffness degradation and stress redistribution on overall fatigue performance. The resulting component and overall fatigue life data are then used to develop a Spatial-Topology and Physics-Informed Cascade Network (STPI-Net). The network uses the spatial configuration and prestress loss characteristics of the tendons as input information to predict fatigue life at critical regions and for the overall structure. The proposed integrated analysis framework combines high-fidelity numerical simulation with data-driven prediction. It provides an approach for fatigue performance evaluation and life prediction of steel–prestressed concrete hybrid wind turbine towers under different prestress relaxation conditions.

2. Fatigue Analysis Methodology

Steel–prestressed concrete hybrid towers are subjected to alternating wind loads throughout their service life, rendering the tower structure highly susceptible to fatigue-related issues. Over the typical design life of 20 years, the cumulative number of stress cycles experienced by the tower can reach 5.29×10^8 , placing the tower response within the high-cycle fatigue regime [26]. To evaluate the fatigue life of the hybrid tower, this section outlines the fundamental theories and methods employed in time-domain fatigue analysis, encompassing the rainflow counting method, S–N curves, mean stress correction, and the linear cumulative damage rule. The flowchart is presented in Figure 1.

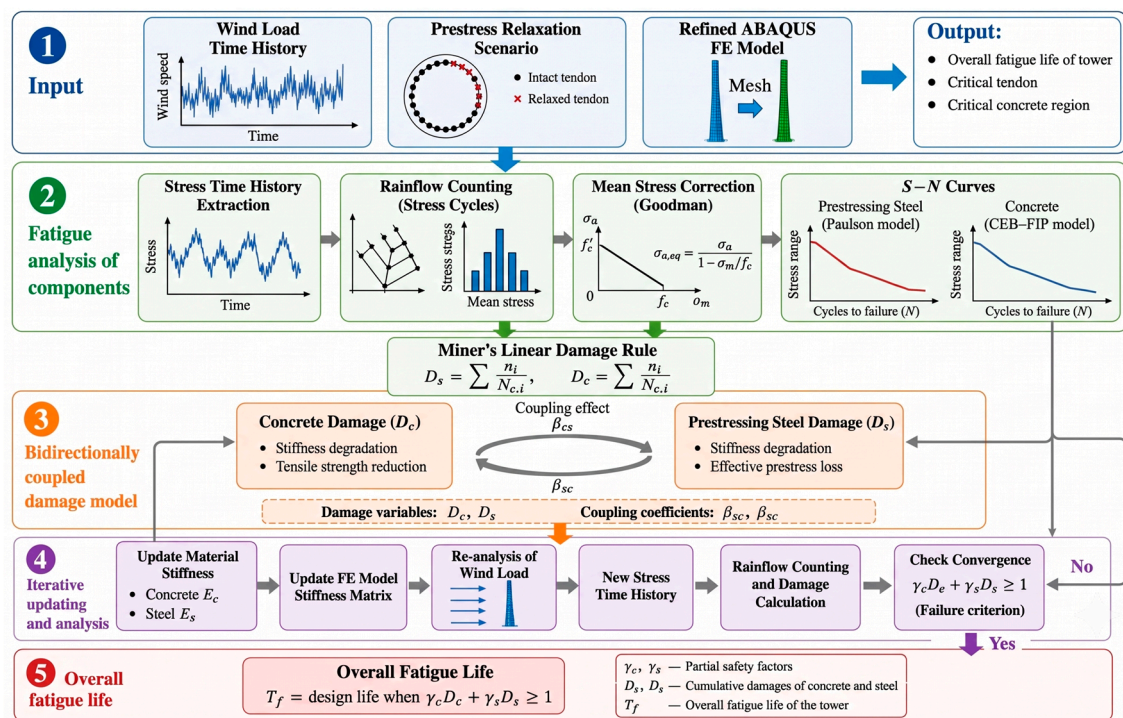


Figure 1. Framework of the proposed bidirectionally coupled fatigue damage assessment.

2.1. Rainflow Counting Method

The rainflow counting method [27] is used to convert each stress time history into a set of fatigue cycles. In this study, the rainflow function in MATLAB R2020b is used to perform the cycle counting. This function follows ASTM E1049-85 [28]. It first identifies the turning points in the stress history, and then extracts closed cycles. Complete cycles are counted as full cycles, while unmatched residuals are counted as half-cycles. For each identified cycle, the stress range or amplitude, mean stress, and cycle count are obtained. These results are subsequently used for mean-stress correction and cumulative fatigue-damage calculation.

2.2. S-N Curves

S-N curves describe the relationship between stress level and fatigue life under constant-amplitude cyclic loading. Because the prestressing tendons and concrete have different fatigue properties, separate S-N relations are used for the two materials.

The fatigue life of the grade 1860 prestressing steel strands is evaluated using the S-N model proposed by Paulson et al. [29]. The model was developed from fatigue-test data for more than 700 prestressing strands and is expressed as

$$\lg N = 13.95 - 3.5 \lg S_r \quad (1)$$

where N denotes the number of stress cycles; S_r is the stress range of the prestressing tendon in MPa.

Since the actual mean stress level of the strands in the present investigation does not satisfy the assumption adopted in the fatigue life formulation, the Goodman linear correction model [30] is introduced to modify the stress as follows:

$$\Delta \sigma_G = \frac{\Delta \sigma}{1 - \frac{S_m}{S_u}} \quad (2)$$

where $\Delta\sigma_G$ is the Goodman-corrected stress range; $\Delta\sigma$ is the original stress range; S_m is the mean stress of the corresponding stress cycle (MPa); and S_u is the ultimate tensile strength of the prestressing strand (MPa).

In a steel–prestressed concrete hybrid tower, the concrete component serves as the primary compression-resisting member, sustaining the vast majority of compressive stresses. Its fatigue performance therefore has an important influence on the safety and durability of the entire structure. Once fatigue damage initiates, crack nucleation and propagation may accelerate structural degradation. Consequently, a systematic assessment of the fatigue life of the concrete portion is essential.

Because concrete fatigue behavior is highly dependent on the stress state, distinct S–N curves must be adopted for the compressive and tensile loading regimes, respectively, as specified in the CEB–FIP Model Code provisions [31]. According to Lohaus et al. [32], the fatigue design provisions of the CEB–FIP Model Code apply to concrete strength classes up to C80. Therefore, the C80 high-strength concrete used in this study is at the upper limit of the specified range.

For concrete under full compression, the S–N curves are expressed as

$$\lg N_1 = 12 + 16S_{c,min} + 8S_{c,min}^2 \quad (3)$$

$$\lg N_2 = 0.2 \lg N_1 (\lg N_1 - 1) \quad (4)$$

$$\lg N_3 = \frac{\lg N_2 (0.3 - 0.375S_{c,min})}{\Delta S_c} \quad (5)$$

The final fatigue life, expressed as $\log N$, is selected according to the following rules:

$$\lg N = \begin{cases} \lg N_1, & \lg N_1 \leq 6 \\ \lg N_2, & \lg N_1 > 6 \text{ and } \Delta S_c \geq 0.3 - 0.375S_{c,min} \\ \lg N_3, & \lg N_1 > 6 \text{ and } \Delta S_c < 0.3 - 0.375S_{c,min} \end{cases} \quad (6)$$

where $\Delta S_{c,min} = \sigma_{c,min}/f_c$ and $\Delta S_c = \sigma_c/f_c$ are the stress level and stress-amplitude ratio.

When the concrete is subjected to cyclic loading that induces tensile stresses, the corresponding S–N curves are categorised into two cases according to the relative magnitude of the tensile stress. These scenarios can be expressed uniformly in the following piecewise form:

$$\lg N = \begin{cases} 9(1 - S_{ct,max}), & \sigma_{ct,max} \leq 0.026 \mid \sigma_{c,max} \mid \\ 12(1 - S_{ct,max}), & \sigma_{ct,max} > 0.026 \mid \sigma_{c,max} \mid \end{cases} \quad (7)$$

where N is the number of cycles; $S_{c,min}$ is the minimum compressive stress level; ΔS_c is the range of stress-level variation; $\sigma_{ct,max}$ is the maximum tensile stress; $\sigma_{c,max}$ is the maximum compressive stress; $S_{c,max}$ is the maximum compressive stress level; and $S_{ct,max}$ is the maximum tensile stress level.

2.3. Linear Cumulative Damage Theory

Under variable-amplitude fatigue loading, structural components experience cycles across various stress ranges, where the total fatigue damage is commonly evaluated via linear cumulative damage. The Palmgren–Miner rule is widely used for this purpose [33]. This theory operates under two core assumptions: (i) the total dissipated work required to cause failure is constant W regardless of the loading history and (ii) each stress level S_i corresponds to a specific constant-amplitude fatigue life N_i . On the basis of these assumptions, when the structure undergoes n_i cycles at the i -th stress level, the incremental damage

incurred at this level is expressed as the ratio of the applied cycles to the corresponding fatigue life:

$$\frac{W_i}{W} = \frac{n_i}{N_i} \quad (8)$$

where W_i denotes the work dissipated at the i -th stress level.

Defining the fatigue damage D as the ratio of the accumulated cycle count to the corresponding fatigue life, for a single constant-amplitude stress level we have

$$D = \frac{n}{N} \quad (9)$$

For variable-amplitude loading histories, the total damage is given by the linear summation of the damage contributions from each stress-range level:

$$D = \sum_i \frac{n_i}{N_i} \quad (10)$$

Failure of the structure is considered to occur when $D \geq 1$.

2.4. Bidirectionally Coupled Damage Model

A bidirectionally coupled damage model is established to describe the interaction between concrete and prestressing tendon damage under cyclic loading. Specifically, concrete cracking reduces structural stiffness and redistributes internal forces [34], raising stress ranges in prestressing tendons. Meanwhile, tendon fatigue damage weakens concrete precompression reserve. To formalize this bidirectional deterioration process, the concrete damage variable D_c and the prestressing tendon damage variable D_s are defined, where 0 represents the intact state, and 1 represents fatigue failure of the corresponding component.

A scalar stiffness-degradation relation is adopted to update the material stiffness:

$$E_c = E_{c0}(1 - D_c) \quad (11)$$

$$E_s = E_{s0}(1 - D_s) \quad (12)$$

where E_{c0} and E_{s0} are the initial elastic moduli of concrete and prestressing tendons, respectively

Within each service-time increment, the damage evolution of each material depends on its stress state and the damage state of the other material. To clarify this process, two state-dependent coupling factors β_{cs} and β_{sc} are introduced. β_{cs} is the acceleration factor of tendon damage on concrete deterioration, and β_{sc} denotes the acceleration factor of concrete damage on tendon deterioration. The damage growth rates are written as

$$\frac{dD_c}{dt} = \frac{1}{T_c^{(0)}}(1 + \beta_{cs}D_s) \quad (13)$$

$$\frac{dD_s}{dt} = \frac{1}{T_s^{(0)}}(1 + \beta_{sc}D_c) \quad (14)$$

where $T_c^{(0)}$ and $T_s^{(0)}$ represent the component fatigue lives calculated for the current prestress-loss case; t denotes the accumulated service time.

β_{cs}, β_{sc} are defined as

$$\beta_{cs} = \left(\frac{\Delta\sigma_c(D_s)}{\Delta\sigma_{c0}} \right)^{m_c} \quad (15)$$

$$\beta_{sc} = \left(\frac{\Delta\sigma_s(D_c)}{\Delta\sigma_{s0}} \right)^{m_s} \quad (16)$$

where $\Delta\sigma_c(D_s)$ denotes the concrete fatigue stress range accounting for the current damage state of the prestressing tendon; $\Delta\sigma_s(D_c)$ denotes the tendon fatigue stress range accounting for the current damage state of concrete; $\Delta\sigma_{c0}$ and $\Delta\sigma_{s0}$ represent the fatigue stress ranges of concrete and tendon under the initial undamaged state; m_c and m_s are the slope parameters of the fatigue S–N curves for concrete and prestressing tendon.

The overall fatigue-life limit is defined using the following weighted damage criterion:

$$\gamma_c D_c + \gamma_s D_s \geq 1 \quad (17)$$

where γ_c and γ_s represent structural weighting coefficients that reflect the respective contributions of the concrete cross-section and the tendon layout to the overall overturning resistance. $\gamma_s/\gamma_c = \frac{(E_s A_s)}{(E_c A_c)}$, $A_s/A_c \approx 1.9\%$ is the cross-sectional area ratio of the tendon to concrete.

Based on the above theories, the specific procedure is as follows:

- (1) Dynamic analysis is carried out on the refined finite element model under self-weight, prestress, and multi-speed wind time histories. Rainflow counting, material S–N curves, Goodman correction, and Miner's rule are adopted to solve single-cycle incremental damage, ΔD_c , ΔD_s .
- (2) The elastic moduli are updated via Equations (11) and (12) with current damage, $D_c^{(n)}$, $D_s^{(n)}$. The updated material properties are assigned in the ABAQUS model to obtain the revised stiffness distribution.
- (3) The finite element analysis is then repeated with updated stiffness. Concrete cracking raises tendon stress ranges, while tendon prestress loss accelerates concrete tensile damage. The cycle incremental damage is superposed following Equations (13) and (14), and the total damages are updated as

$$D_c^{(n+1)} = D_c^{(n)} + \Delta D_c \quad (18)$$

$$D_s^{(n+1)} = D_s^{(n)} + \Delta D_s \quad (19)$$

- (4) Substitute updated $D_c^{(n+1)}$ and $D_s^{(n+1)}$ into Equation (17). If $\gamma_c D_c + \gamma_s D_s < 1$, repeat Steps (2) and (3) for the next cycle. Iteration terminates once $\gamma_c D_c + \gamma_s D_s \geq 1$ and the total iterative cycles correspond to the overall tower fatigue life with mutual material deterioration considered.

This stiffness-feedback iteration effectively couples the dual-material damage evolution by recycling the pre-calculated stress spectra and baseline damage data.

3. Finite Element Model of the Hybrid Tower

3.1. Finite Element Modeling

The model adopted in this study is based on an in-service 4.55 MW three-bladed steel–prestressed concrete hybrid wind turbine tower. The key design parameters of this wind tower are summarised in Table 1. As shown in Figure 2, the concrete section comprises 35 precast segments, numbered Part 1 to Part 35 from bottom to top, together with the door-opening region A0. Reinforcing bars are embedded within the concrete, and the precast concrete segments are tied together. The prestressing strands are arranged as external tendons, connecting the tower foundation to the transition piece. In terms of element selection, both the concrete and steel components are discretised using eight-node linear hexahedral reduced-integration elements (C3D8R). The reinforcing bars and prestressing tendons are represented by two-node linear three-dimensional truss elements (T3D2). The concrete grade is C80, and its nonlinear behaviour is described by the Concrete

Damaged Plasticity (CDP) model. The reinforcing steel is HRB400, the steel tower section is Q355, and the prestressing strands have an ultimate tensile strength of 1860 MPa. A structural damping ratio of 1% is adopted in the dynamic analysis to represent the energy dissipation of the wind turbine tower. Rayleigh damping is defined using the first two natural frequencies obtained from the modal analysis.

Table 1. Tower design parameters.

Number of Blades	Blade Radius [m]	Rated Power [MW]	Tower Height [m]	Maximum Rotor Speed [rpm] [r/min]	Weight of Wind Turbine and Blades [ton]
3	92.5	4.55	157.63	8.5	250

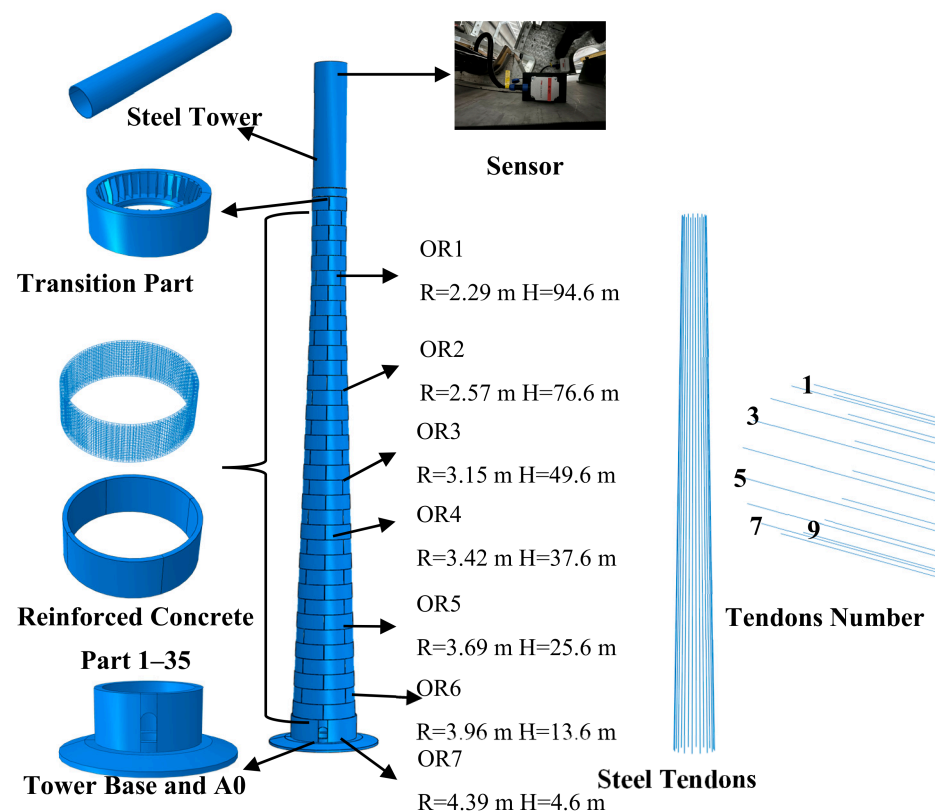


Figure 2. Finite-element model.

3.2. Load Configuration

The loads acting on the tower include structural self-weight, the nacelle and rotor mass, rotor thrust, hub torque, prestress, and along-wind load on the tower.

3.2.1. Dead Load and Prestress Application

In the finite element model, the structural self-weight is calculated from the material densities using a gravitational acceleration of 9.8 m/s^2 . The nacelle and rotor masses are represented by concentrated masses at the tower top. Prestressing loads are applied using the equivalent temperature drop method. Sixteen external prestressing tendons are arranged inside the concrete tower section, each consisting of 18 high-strength low-relaxation strands, with an initial prestress of 1260 MPa. The equivalent temperature change is expressed as

$$\Delta T = -\frac{\sigma_p}{\alpha E_s} \quad (20)$$

where ΔT is the equivalent temperature change; σ_p is the initial prestress of the strand; $E_s = 200$ GPa is the elastic modulus of the prestressing strand; $\alpha = 1.2 \times 10^{-5}$ is the coefficient of thermal expansion of the strand.

3.2.2. Stochastic Wind Field Simulation

The total duration of wind field simulation in this study is set to 600 s, with a sampling frequency of 20 Hz. The wind speed is decomposed into mean and fluctuating components. The time-domain stochastic wind field simulation method is adopted to generate wind-speed time histories:

$$V(z, t) = \bar{V}(z) + v'(z, t) \quad (21)$$

where $V(z, t)$ is the instantaneous wind speed at height z ; $\bar{V}(z)$ is the mean wind speed; $v'(z, t)$ is the fluctuating wind speed.

$\bar{V}(z)$ along the height direction is represented by an exponential wind profile:

$$\bar{V}(z) = V_{10} \left(\frac{z}{10} \right)^\alpha \quad (22)$$

where V_{10} is the reference wind speed at 10 m height; z is the height of the calculation position; α is the terrain roughness exponent.

Since the fluctuating wind speed process is stochastic, the harmonic superposition method is employed to generate the fluctuating wind speed time history. The fluctuating wind speed is expressed as the superposition of multiple harmonic components with different frequencies:

$$v'(z, t) = \sum_{m=1}^M \sqrt{2S_v(f_m)\Delta f} \cos(2\pi f_m t + \phi_m) \quad (23)$$

where $S_v(f_m)$ is the wind speed power spectral density; f_m is the m discrete frequency; Δf is the frequency interval; M is the number of discrete frequencies; t is the time variable; ϕ_m is the random phase angle. The random phase satisfies $\phi_m \in [0, 2\pi]$ and is randomly generated following a uniform distribution.

The simulated spectra are compared with the target Kaimal spectrum to verify the frequency characteristics of the generated wind field.

$$S_u(f) = \frac{200u_*^2 z/U}{(1 + 50fz/U)^{5/3}} \quad (24)$$

where $S_u(f)$ is the longitudinal turbulence power spectral density; $u_* = \frac{0.4U}{\ln(z/z_0)}$ is the friction velocity, where z_0 is the ground roughness length; U is the mean wind speed; and f is the frequency variable.

Fluctuating wind speeds at different heights are spatially correlated. To account for this effect, the spatial coherence function is adopted in this study to describe the wind speed correlation characteristics between different spatial positions. The spatial coherence function is defined as

$$Coh_{ij}(f) = \exp\left(-\frac{C_z f |z_i - z_j|}{\bar{V}_{ij}}\right) \quad (25)$$

where $Coh_{ij}(f)$ is the spatial coherence coefficient; i and j denote two spatial positions, respectively; z_i and z_j are the height coordinates of the corresponding positions; $\bar{V}_{ij}$ is the mean wind speed at the two positions; $C_z = 10$ is the spatial coherence decay coefficient.

The cross-power spectral density considering spatial correlation is expressed as

$$S_{ij}(f) = \sqrt{S_i(f)S_j(f)}Coh_{ij}(f) \quad (26)$$

where $S_i(f)$ and $S_j(f)$ i -th and j -th spatial positions, respectively. Based on the obtained cross-power spectral matrix, Cholesky decomposition is performed to obtain the frequency-domain components of the stochastic wind speed with spatial correlation, and the fluctuating wind speed time histories at each spatial position are obtained through inverse Fourier transform.

3.2.3. Rotor Aerodynamic Force

The time-varying aerodynamic loads on the rotor are calculated using blade element momentum (BEM) theory. At each time step, the local wind speed is obtained from the stochastic wind field described in Section 3.2.2. The relative wind speed and inflow angle of each blade element are determined from the local wind speed and blade rotational speed.

The blade is divided into several blade elements along the spanwise direction. For a blade element at radius r , the thrust and torque increments are expressed as

$$dT = \frac{1}{2}\rho W^2 BcC_n dr \quad (27)$$

$$dQ = \frac{1}{2}\rho W^2 BcC_t r dr \quad (28)$$

where dT and dQ are the thrust and torque increments of the blade element; $\rho = 1.31 \text{ kg/m}^3$ is the air density; W is the relative wind speed at the blade element; $B = 3$ is the number of blades; c is the chord length of the blade element; C_n and C_t are the normal and tangential aerodynamic force coefficients.

The aerodynamic force coefficients are obtained from the lift and drag characteristics of the blade airfoil:

$$C_n = C_L \cos \phi + C_D \sin \phi \quad (29)$$

$$C_t = C_L \sin \phi - C_D \cos \phi \quad (30)$$

where C_L and C_D are the lift and drag coefficients of the airfoil, respectively; and ϕ is the inflow angle. The lift and drag coefficients are determined from the airfoil data according to the local angle of attack.

The total rotor thrust and torque are obtained by integrating the blade-element contributions along the blade span:

$$T(t) = \int_{r_h}^R dT \quad (31)$$

$$Q(t) = \int_{r_h}^R dQ \quad (32)$$

where r_h is the hub radius; and R is the blade radius. The resulting thrust and torque time histories are applied at the hub as a concentrated force and torsional moment, respectively.

3.2.4. Wind Load on the Tower

The along-wind load acting directly on the tower is calculated using the quasi-steady method:

$$F_w(z, t) = \frac{1}{2}\rho\mu_s A(z)V^2(z, t) \quad (33)$$

where $F_w(z, t)$ is the wind load on the tower at height z ; ρ is the air density, taken as $\rho = 1.31 \text{ kg/m}^3$; μ_s is the tower shape coefficient, taken as $\mu_s = 0.9$; and $A(z)$ is the windward area at the corresponding height; $V(z, t)$ is the local instantaneous wind speed.

For the rotor-affected region, the local inflow wind speed is corrected by considering the velocity variation induced by rotor rotation:

$$V_w(z, t) = V_\infty(z, t)(1 - \Delta V) \quad (34)$$

where $V_w(z, t)$ is the corrected local wind speed; $V_\infty(z, t)$ is the inflow wind speed unaffected by the rotor; and ΔV is the velocity deficit coefficient induced by the rotor. The corrected local wind speed is used to calculate the wind load on the tower surface in the rotor-affected region.

3.3. Prestressing Conditions

As illustrated in Figure 2, the 16 prestressing tendons are sequentially numbered from 1 to 16. Tendon No. 1 is aligned with the incoming wind direction (i.e., at 0° relative to the mean incoming wind direction), while Tendon No. 9 is positioned on the opposite side (at 180°). The remaining tendons are numbered clockwise.

In this study, two categories of prestress-loss cases are designed: single-tendon relaxation and multi-tendon relaxation. The detailed configurations are presented in Table 2. BL denotes the case without prestress loss.

Table 2. Prestressing tendon relaxation cases.

Case ID	Number of Relaxed Tendons	Relaxed Tendon IDs	Angle Between Relaxed Tendons and Wind Direction ($^\circ$)
BL	0	N/A	N/A
SR1	1	1	0
SR2	1	3	45
SR3	1	5	90
SR4	1	7	135
SR5	1	9	180
MR1	3	1, 2, 16	0, 22.5, -22.5
MR2	3	1, 2, 3	0, 22.5, 45
MR3	3	1, 3, 15	0, 45, -45
MR4	3	1, 3, 5	0, 45, 90

For the single-tendon relaxation cases (SR1–SR5), prestress loss levels of 20%, 40%, 60%, and 80% are adopted, corresponding to equivalent temperature drops of -420°C , -315°C , -210°C , and -105°C (with corresponding stresses of 1008, 756, 504, and 252 MPa), respectively. For the multi-tendon relaxation cases (MR1–MR4), the same loss levels are applied to the relaxed tendons. The 60% and 80% prestress-loss cases are introduced as severe parametric scenarios rather than as loss levels expected from normal tendon relaxation during service. These cases are used to examine fatigue sensitivity and identify critical degradation under prestress loss. At 80% prestress loss, the initial residual tendon stress is approximately 252 MPa, while the minimum stress during the simulated case remains positive at 223.37 MPa. This result confirms that the tendons remain engaged throughout the investigated loading range. Accordingly, the high-loss cases are used to examine the effects of substantial prestress reduction on coupled damage evolution and overall fatigue life while the tendons remain effectively engaged.

In the case setup, prestress loss levels are represented by several discrete states to evaluate the influence of different degradation degrees on the fatigue performance of

hybrid towers. However, during actual service, strand relaxation exhibits significant time dependence, and the prestress loss accumulates gradually over time in a nonlinear manner. To further clarify the correspondence between the preset prestress loss levels and the actual service degradation process, this study analyzes the time evolution characteristics of strand prestress loss over a two-year period. Referring to an existing time-dependent prestress loss model [35], the normalized form is adopted to calculate the variation in loss rate with time:

$$\frac{\Delta\sigma_{pr}}{\sigma_{pi}} = 0.66\rho_{1000}e^{9.1\mu}\left(\frac{t}{1000}\right)^{0.75(1-\mu)}10^{-5} \quad (35)$$

where $\Delta\sigma_{pr}$ is the absolute value of relaxation stress; σ_{pi} is the prestress after short-term post-tensioning loss; ρ_{1000} is the strand relaxation percentage at 1000 h and 20 °C, taken as $\rho_{1000} = 2.5\%$ for low relaxation strands; t is the time after tensioning in hours; $\mu = \sigma_{pi}/f_{ptk}$; and f_{ptk} is the characteristic value of the ultimate tensile strength of the prestressing tendon.

Figure 3 shows the prestress evolution of windward tendons 1 and 3 and leeward tendons 7 and 9 over a two-year period. Under normal conditions, the loss of strands within two years is small, while under relaxation conditions, strands at all positions exhibit a nonlinear increase in prestress loss, with the loss mainly concentrated in the early service stage and then gradually decreasing and stabilizing. In terms of spatial position, the maximum monthly loss rates of windward tendons 1 and 3 are 7.79% and 8.12%, respectively, while those of leeward tendons 7 and 9 reach 9.67% and 9.94%, respectively. Compared with the windward side, the leeward strands have a higher initial loss rate and a higher initial rate of prestress loss, resulting in non-uniform changes in the cross-sectional prestress distribution and further affecting the fatigue damage development of the hybrid tower. Therefore, the time accumulation characteristics and spatial differences in prestress loss should be considered in fatigue performance evaluation.

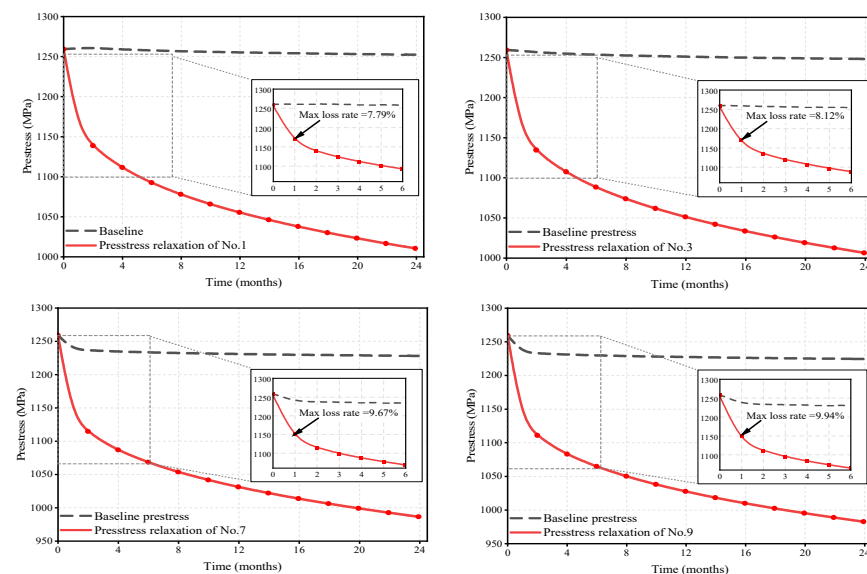


Figure 3. Time-dependent evolution of tendon prestress under relaxation.

3.4. Wind Speed Conditions

In all cases, the magnitude of the wind load acting on the tower is adjusted by varying the wind speed to simulate the dynamic responses under different velocities. A probability weighting approach is then used to provide a reasonable assessment of cumulative fatigue damage over the design service life. In long-term fatigue evaluation of wind turbines, the long-term wind speed distribution is typically fitted using a Weibull distribu-

tion [36]. In this study, the two-parameter Weibull distribution is employed, as given by Equation (36) [37] and Figure 4.

$$f(v; k, \lambda) = \frac{k}{\lambda} \left(\frac{v}{\lambda} \right)^{k-1} \exp \left[- \left(\frac{v}{\lambda} \right)^k \right] \quad (36)$$

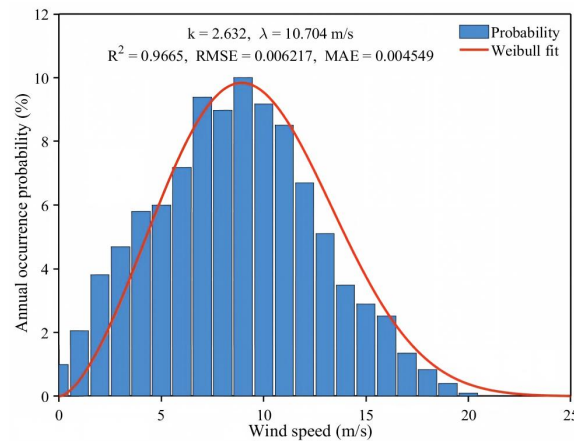


Figure 4. Wind speed frequency distribution.

The corresponding cumulative distribution function (CDF) is:

$$F(v; k, \lambda) = 1 - \exp \left[- \left(\frac{v}{\lambda} \right)^k \right] \quad (37)$$

where v is the wind speed (m/s); k is the dimensionless shape parameter; and λ is the scale parameter (m/s).

Based on the wind-resource records for the project site, the fit to the available wind-resource data gives $k = 2.632$, $\lambda = 10.704$ m/s, as shown in Figure 4. The sensitivity of the fit is examined by varying k and λ individually by $\pm 5\%$. As shown in Table 3, R^2 remains above 0.94, while RMSE and MAE remain low. Therefore, the Weibull distribution still adequately describes the wind speed probability characteristics within a $\pm 5\%$ parameter fluctuation range.

Table 3. Sensitivity of goodness-of-fit metrics to Weibull parameter variations.

Case	k	λ	R^2	RMSE	MAE
Baseline	2.632	10.704	0.96649	0.00622	0.00455
$k - 5\%$	2.500	10.704	0.96199	0.00665	0.00463
$k + 5\%$	2.764	10.704	0.95454	0.00724	0.00522
$\lambda - 5\%$	2.632	10.169	0.95130	0.00750	0.00503
$\lambda + 5\%$	2.632	11.239	0.94531	0.00794	0.00628

The cut-in, rated, and cut-out wind speeds of the turbine are 3, 11.9, and 25 m/s, respectively. Considering these operating limits and the long-term wind-speed distribution, six representative wind speeds of 3, 6, 9, 12, 15, and 18 m/s are selected. The probability of each wind-speed interval is obtained by integrating the Weibull probability density function. To preserve the total probability in the six-state discretization, the lower and upper tails of the distribution are assigned to the nearest representative wind-speed conditions. The resulting intervals and probabilities are listed in Table 4. Annual occurrence

probability P_j for each wind speed is obtained by integrating the Weibull PDF over the corresponding interval:

$$P_j = \int_{v_{j,l}}^{v_{j,h}} f(v; k, \lambda) dv = \exp \left[- \left(\frac{v_{j,l}}{\lambda} \right)^k \right] - \exp \left[- \left(\frac{v_{j,h}}{\lambda} \right)^k \right] \quad (38)$$

where $v_{j,l}$ and $v_{j,h}$ are the lower and upper bounds, respectively, of the j -th wind-speed bin. The annual weighting factors for each wind speed are summarised in Table 4.

Table 4. Annual occurrence probability for each wind-speed bin.

Wind Speed (m/s)	Wind Speed Bin (m/s)	Annual Occurrence Probability
3	$v \leq 4.5$	0.0965
6	$4.5 < v \leq 7.5$	0.2279
9	$7.5 < v \leq 10.5$	0.2968
12	$10.5 < v \leq 13.5$	0.2189
15	$13.5 < v \leq 16.5$	0.1257
18	$v > 16.5$	0.0342

Long-term fatigue assessment of wind turbine structures usually requires short-duration simulation or measurement results to be extrapolated to the design period. Probability-based extrapolation methods have been widely used to estimate long-term fatigue damage from short-duration load responses [38]. In this study, a 600 s dynamic time-history analysis is performed for each representative wind speed, and the corresponding stress cycles are obtained using the rainflow counting method. The counted cycles are then scaled to one year according to the ratio of the annual duration to the simulation duration. Based on the Palmgren–Miner rule described in Section 2.3, the annual fatigue damage associated with the j -th representative wind speed is calculated as

$$D_j = \frac{n_j}{N_j} \quad (39)$$

where D_j is the fatigue damage corresponding to the j -th representative wind speed acting continuously for one year; N_j is the theoretical number of fatigue cycles corresponding to the j -th representative wind speed; $n_j = n'_j \cdot \frac{t_y}{600}$, where n_j is the number of cycles converted under the condition of continuous action throughout the year for the j -th representative wind speed, n'_j is the number of cycles obtained by rainflow counting from the 600 s simulated time history under the j -th representative wind speed, and t_y is the total duration of one year (s).

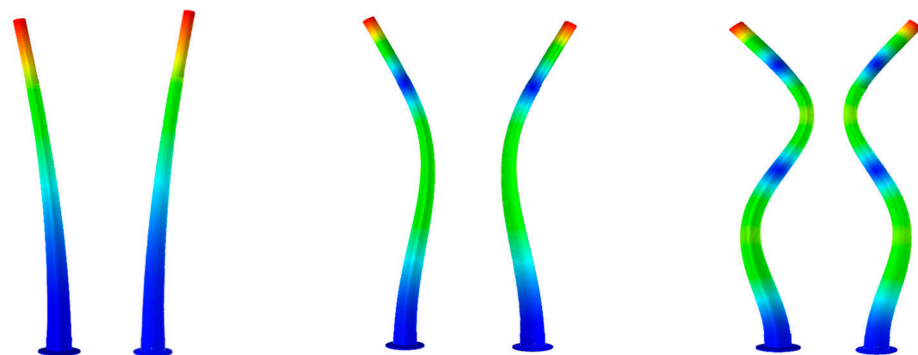
The final annual cumulative fatigue damage is expressed as [39]

$$D = \sum_{j=1}^6 P_j D_j \quad (40)$$

3.5. Model Validation

Using the established finite-element model, the modal characteristics of the hybrid tower were first obtained, as shown in Figure 5. Measured modal characteristics have been widely used to validate and update finite element models of wind turbine towers [40]. For the investigated tower, the maximum rotor speed of the wind turbine is 8.5 r/min, from which the rotational frequency of the rotor is computed as $f_1 = 8.5/60 = 0.1417$ Hz. As the turbine is equipped with three blades, the blade-passing frequency is $f_3 = 3 \times f_1 = 0.425$ Hz. To avoid resonance, the rotor rotational frequency must fall outside the interval of 0.1275–0.1559

Hz, and the blade-passing frequency must avoid the range of 0.3825–0.4675 Hz. As can be observed from Figure 5, all the computed natural frequencies lie outside these critical resonance bands.



The first three natural frequencies in the x- and y-directions are listed as follows:

First mode: 0.2101 Hz (x)/0.2112 Hz (y); 2nd mode: 1.1492 Hz (x)/1.1483 Hz (y);

Third mode: 2.9578 Hz (x)/2.9631 Hz (y).

Figure 5. Model analysis results.

To further assess the model using independent field data, one-year monitoring results from the in-service tower were introduced. Horizontal vibration was measured in two orthogonal directions using JBA22 dual-axis accelerometers. Tower-top displacement was estimated from the inclination recorded by JQIN-04 biaxial inclinometers using the small-angle geometric relation for the tower. The mean tensile forces of Tendons Nos. 1, 3, 5, 7, and 9 were also compared with the corresponding finite element results. The main sensor specifications are listed in Table 5. The main sensor specifications are listed in Table 5.

Table 5. Main Specifications of the Field Monitoring Sensors.

Parameter	JBA22 Accelerometer	JQIN-04 Inclinometer
Measurement	Horizontal vibration	Tower inclination
Axes	Two horizontal axes	X and Y
Measurement range	± 2 g	$\pm 15^\circ$
Accuracy	Linearity $< 0.1\%$	Dynamic accuracy: 0.1°
Frequency response/Resolution	0–120 Hz	0.001°

Field monitoring showed that the dominant vibration frequencies in the x- and y-directions were both approximately 0.224 Hz. The first-mode frequencies obtained from the finite element model were 0.2101 Hz and 0.2112 Hz, corresponding to relative differences of 6.21% and 5.71%, respectively. For the absolute tower-top displacement, the relative differences in the mean, standard deviation, and maximum values were 4.61%, 3.94%, and 7.76%, respectively. The relative differences in mean tensile force for Tendons Nos. 1, 3, 5, 7, and 9 ranged from 1.24% to 7.04%. These comparisons are summarized in Table 6; the results support the ability of the finite element model to reproduce the global dynamic characteristics under the compared conditions.

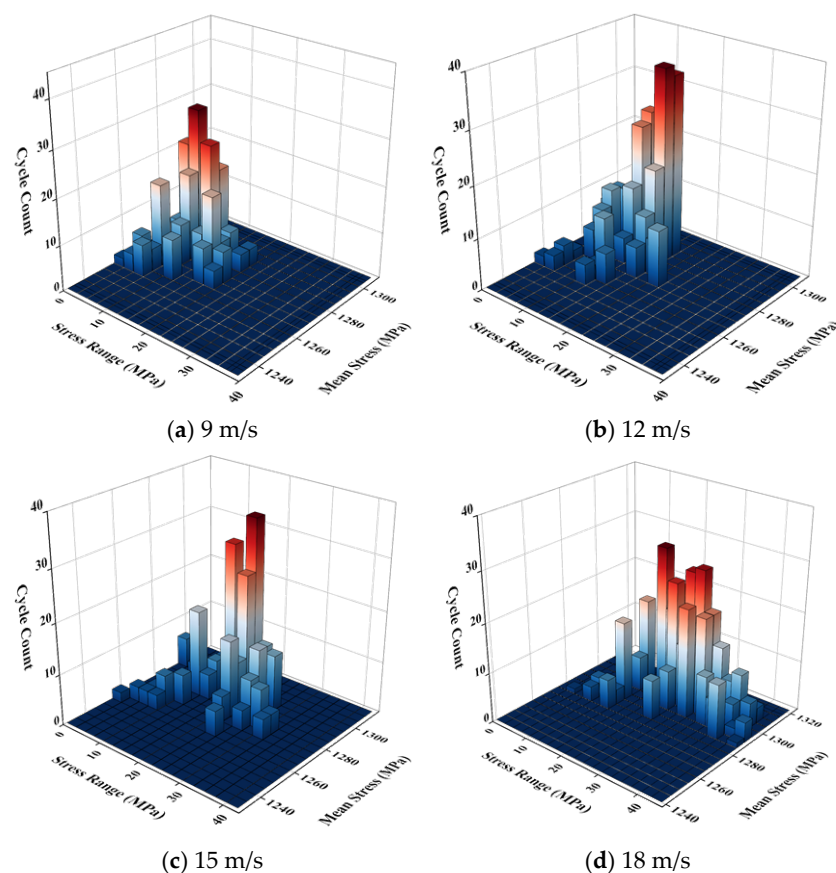
Table 6. Comparison of field measurements and numerical results for global model validation.

Response Type	Parameter	Field Measurement	FE Result	Corrected Difference (%)
Dynamic characteristic	1st mode (x-dir.)	0.224 Hz	0.2101 Hz	6.21
Dynamic characteristic	1st mode (y-dir.)	0.224 Hz	0.2112 Hz	5.71
Tower-top displacement	Mean	565.92 mm	539.8 mm	4.61
Tower-top displacement	Standard deviation	271.94 mm	282.65 mm	3.94
Tower-top displacement	Maximum	862.99 mm	929.96 mm	7.76
Tendon No. 1 force	Mean	3288.33 KN	3459.83 KN	5.22
Tendon No. 3 force	Mean	3265.245 KN	3219.61 KN	1.40
Tendon No. 5 force	Mean	3206.25 KN	3166.58 KN	1.24
Tendon No. 7 force	Mean	3149.82 KN	3080.77 KN	2.19
Tendon No. 9 force	Mean	3126.735 KN	2906.59 KN	7.04

4. Fatigue Life of Prestressing Tendons

4.1. Baseline Fatigue Performance of Prestressing Tendons

The fatigue performance of the 16 prestressing tendons distributed across the tower cross-section was first evaluated under the baseline (BL) case, which represents the long-term wind loading condition. Tendon No. 1 was selected as an example to analyze the stress variation characteristics. The axial stress time history was extracted, and the rainflow counting method was applied to obtain the stress cycles, including the stress range and mean stress. Figure 6a–d present the stress spectra under wind speeds of 9, 12, 15, and 18 m/s.

**Figure 6.** Load spectrum of Tendon No. 1.

Overall, the stress ranges at different wind speeds are mainly concentrated below 40 MPa, while the mean stresses are distributed within 1240–1320 MPa. At relatively low wind speeds (9 and 12 m/s), most stress ranges remain below 20 MPa. With increasing wind speed, the stress range distribution gradually shifts toward higher values. This trend is mainly related to the increase in aerodynamic thrust and turbulence intensity under higher wind speeds, which leads to larger cyclic loads transferred from the rotor to the tower structure. As a result, the dynamic stress range of the prestressing tendons increases. The mean stress also shows a slight increase with wind speed. However, its variation is much smaller than that of the stress range. This is because the initial tensile stress of the prestressing tendon is relatively high, and the additional mean stress caused by wind loading has a limited influence compared with the initial stress level.

Based on the stress spectra, tendon fatigue life is calculated using the framework described in Section 2. Taking the 12 m/s wind speed as an example, the stress range and mean stress of each stress cycle are first obtained by rainflow counting. The stress ranges are then corrected using the Goodman relation in Equation (2). Based on the fatigue damage equivalence principle [41], the corrected stress spectrum is converted into an equivalent stress range of 16.89 MPa. Substituting this value into Equation (1) gives the corresponding theoretical number of fatigue cycles, where $N_j = 4.50 \times 10^9$ cycles. The number of cycles obtained from the 600 s stress history is converted into the annual cycle number. The annual fatigue damage under continuous operation at the j -th representative wind speed is calculated using Equation (39), yielding $D_j = 0.03335$. The same procedure is applied to the other representative wind speeds. Finally, the contributions of all wind-speed intervals are accumulated using Equation (40).

Table 7 summarizes the equivalent stress range, equivalent fatigue life cycles, and annual damage contribution at each wind speed for Tendon No. 1 (baseline case). The total annual damage for Tendon No. 1 is 2.504×10^{-2} , corresponding to a fatigue life of 39.9 years.

Table 7. Fatigue life and annual damage of Tendon No. 1.

Wind Speed (m/s)	Equivalent Stress Range (MPa)	Equivalent Fatigue Life (Cycles)	Annual Damage Contribution
3.0	1.07	7.03×10^{13}	3.28×10^{-8}
6.0	4.19	5.92×10^{11}	1.25×10^{-5}
9.0	9.86	2.96×10^{10}	1.15×10^{-3}
12.0	16.89	4.5×10^9	7.30×10^{-3}
15.0	26.35	9.49×10^8	9.46×10^{-3}
18.0	37.03	2.88×10^8	7.11×10^{-3}
Total annual damage: 2.504×10^{-2}		Fatigue life: 39.9 years	

Figure 7 presents the fatigue-life distribution of the prestressing tendons under the baseline condition. The predicted lives of all tendons exceed the 20-year design service life, but a clear spatial variation is observed around the tower cross-section. The windward and leeward tendons, including Tendons Nos. 1, 2, 8, 9, 10, and 16, have relatively short fatigue lives. By contrast, the lateral tendons, such as Tendons Nos. 5 and 13, have much longer lives.

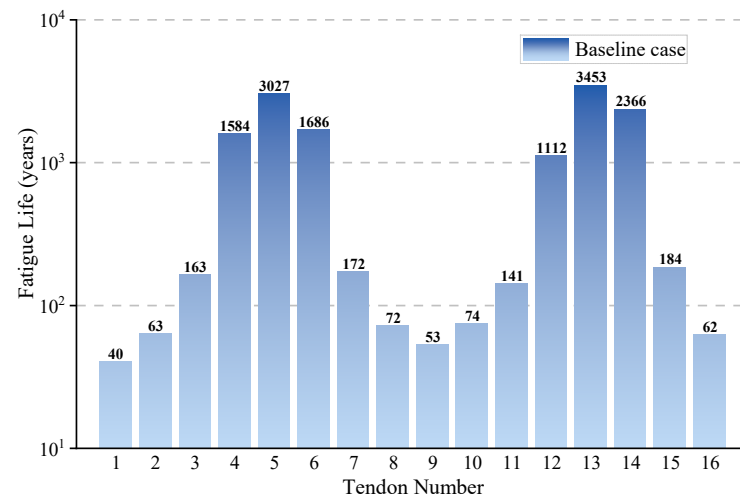


Figure 7. Fatigue life of each prestressing tendon under the baseline case.

This spatial variation is mainly controlled by the tendon position relative to the bending neutral axis. The windward and leeward tendons are farther from the neutral axis and therefore experience larger stress ranges under along-wind bending. The lateral tendons are closer to the neutral region and experience smaller stress variations. The windward and leeward tendons are therefore identified as the fatigue-critical tendons under the baseline condition. Their locations are used in the subsequent analysis of tendon relaxation.

4.2. Prestressing Tendon Fatigue Response Under Tendon Relaxation

Prestress relaxation changes the stress state of the tower and affects the fatigue performance of prestressing tendons. In this section, both single-tendon and three-tendon relaxation cases are considered to evaluate the effects of relaxation position and prestress loss level on tendon fatigue life.

Figure 8a–e shows the results for the single-tendon relaxation cases. The fatigue lives of most tendons decrease as the prestress loss level increases. However, the extent of this reduction also depends on the relaxation position. SR1 and SR2, which represent relaxation near the windward side, cause greater reductions than SR3–SR5. At prestress loss levels above 40%, the lives of some tendons in SR1 and SR2 fall below the 20-year design service life. The effects of SR3–SR5 are less pronounced, and the tendon fatigue lives generally remain above the design life over the investigated loss range.

Figure 8f–i presents the results for the three-tendon relaxation cases. Compared with single-tendon relaxation, simultaneous relaxation of three tendons causes a broader reduction in fatigue life. Even at a prestress loss level of 20%, the fatigue lives of several tendons in the MR cases are already below the design service life. The effect also depends on the tendon arrangement. MR1 and MR2 produce larger changes in the fatigue-life distribution than MR3 and MR4. In MR1 and MR2, the fatigue lives of most critical tendons are below 30 years, and some decrease to less than 10 years. In contrast, the fatigue lives in MR3 and MR4 generally remain above 10 years.

Although fatigue life generally decreases with increasing prestress loss, some tendons show minor non-monotonic variations at individual loss levels. These local variations result from the redistribution of prestressing force, which can reduce the stress range at specific tendon locations. However, they do not change the overall reduction in tendon fatigue life with increasing prestress loss.

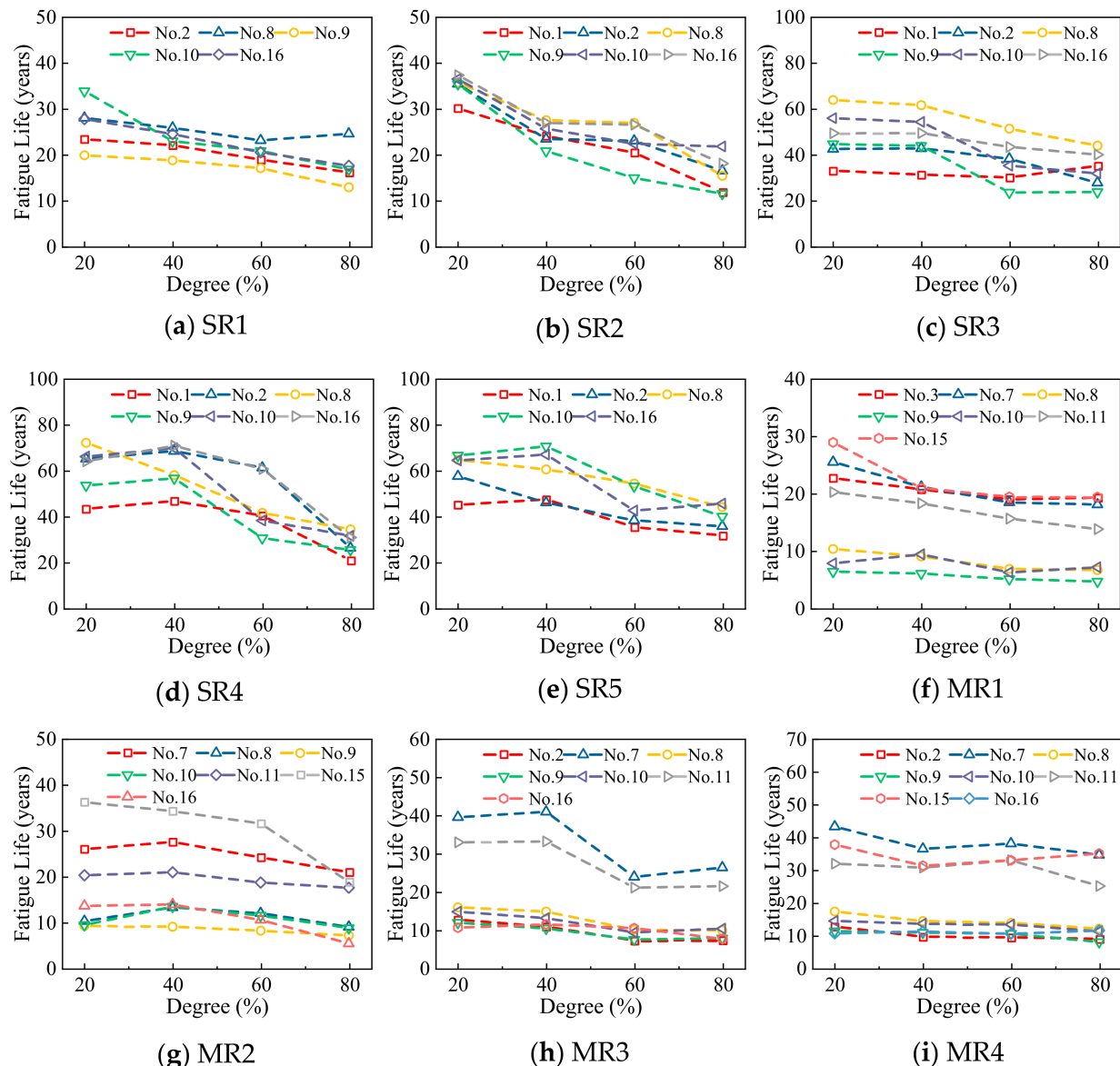


Figure 8. Fatigue life of tendons under various cases.

4.3. Sensitivity Analysis of Tendon Fatigue Life

Because long-term fatigue life also depends on the wind-speed distribution, the sensitivity to the Weibull parameters is further examined. The minimum tendon fatigue life is used as the response indicator for the relaxation cases.

Figure 9 shows the effect of the Weibull parameters on the fatigue life of Tendon No. 1. When $\lambda = 8.0$ m/s, increasing k from 1.0 to 3.0 increases the fatigue life from approximately 20 to 200 years. In contrast, increasing λ from 8.0 to 12.0 m/s reduces the fatigue life for a fixed k . The wide variation indicates that the long-term fatigue prediction is highly dependent on the probability distribution of wind speed, consistent with probabilistic fatigue assessments of wind turbine towers [39]. Therefore, changes in wind-speed probability characteristics can lead to variations in predicted fatigue life.

Figure 10 compares the minimum tendon fatigue life under different relaxation conditions. At 80% relaxation, the minimum lives of SR1 and SR2 decrease to 12.88 and 11.48 years, whereas those of SR3–SR5 remain above the 20-year design life. For the three-tendon cases, the minimum life is already reduced to around 8 years at only 20% relaxation and further decreases to 4.66–8.05 years at 80%. Previous tests on prestressed concrete

beams have shown that prestressing force affects their fatigue S–N relationships [34]. In the present tower, the pronounced life reduction indicates that tendon fatigue is particularly sensitive to simultaneous multi-tendon relaxation.

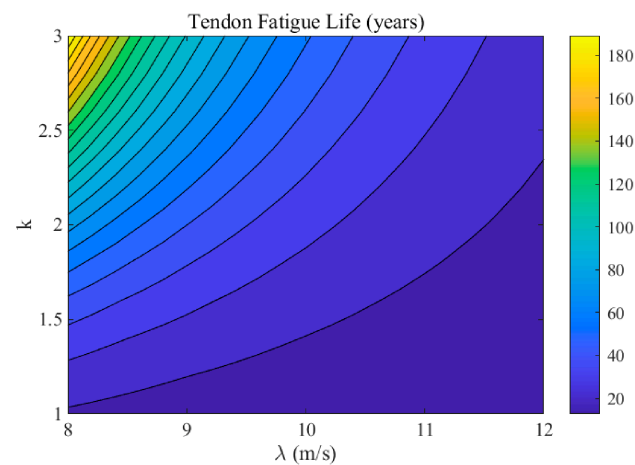


Figure 9. Effect of Weibull parameters on tendon No. 1 fatigue life under the baseline case.

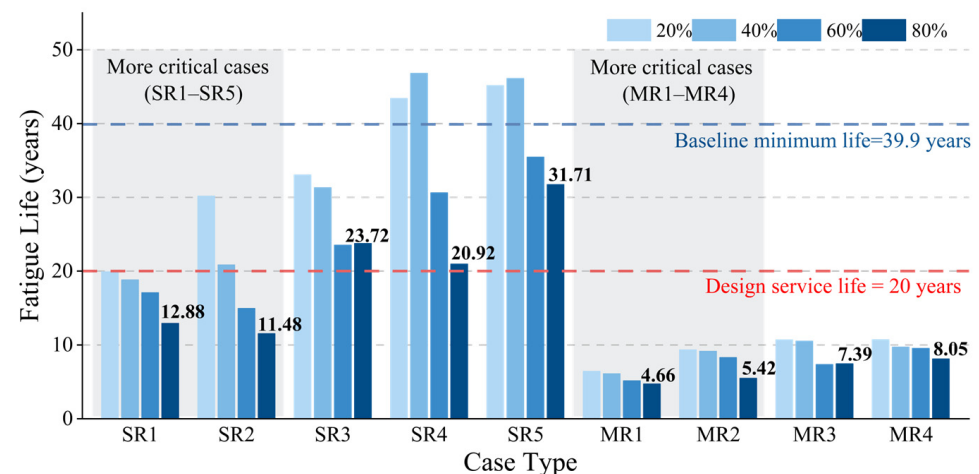


Figure 10. Minimum fatigue life of prestressing tendons.

5. Fatigue Life of Concrete

5.1. Baseline Fatigue Performance of Concrete

In this section, the fatigue performance of the concrete portion of the tower is evaluated under the baseline (BL) case. Seven observation regions, including the A_0 base ring, Part 3, Part 7, Part 11, Part 15, Part 20, and Part 30 (designated OR1 through OR7) were selected. The layout of these points is shown in Figure 2. Each defined region is a zone with a finite spatial extent rather than an idealized geometric point. The regions are selected to cover the main height range of the concrete tower, including geometric discontinuities and connection zones between precast segments.

Taking OR6 as an example, Figure 11 shows the stress spectra under different wind speeds. The stress amplitudes are mainly below 20 MPa, while the mean stresses remain below 35 MPa. As the wind speed increases, the stress-amplitude distribution gradually shifts toward higher values. The 12, 15, and 18 m/s wind-speed conditions contribute more than 90% of the total annual fatigue damage. Although these wind conditions occur less frequently, the larger stress amplitudes generated at higher wind speeds make them the main contributors to concrete fatigue damage. Table 8 summarizes the fatigue damage

contributions corresponding to different wind speeds. The total annual damage at OR6 is 4.09×10^{-2} , corresponding to a fatigue life of 24.46 years.

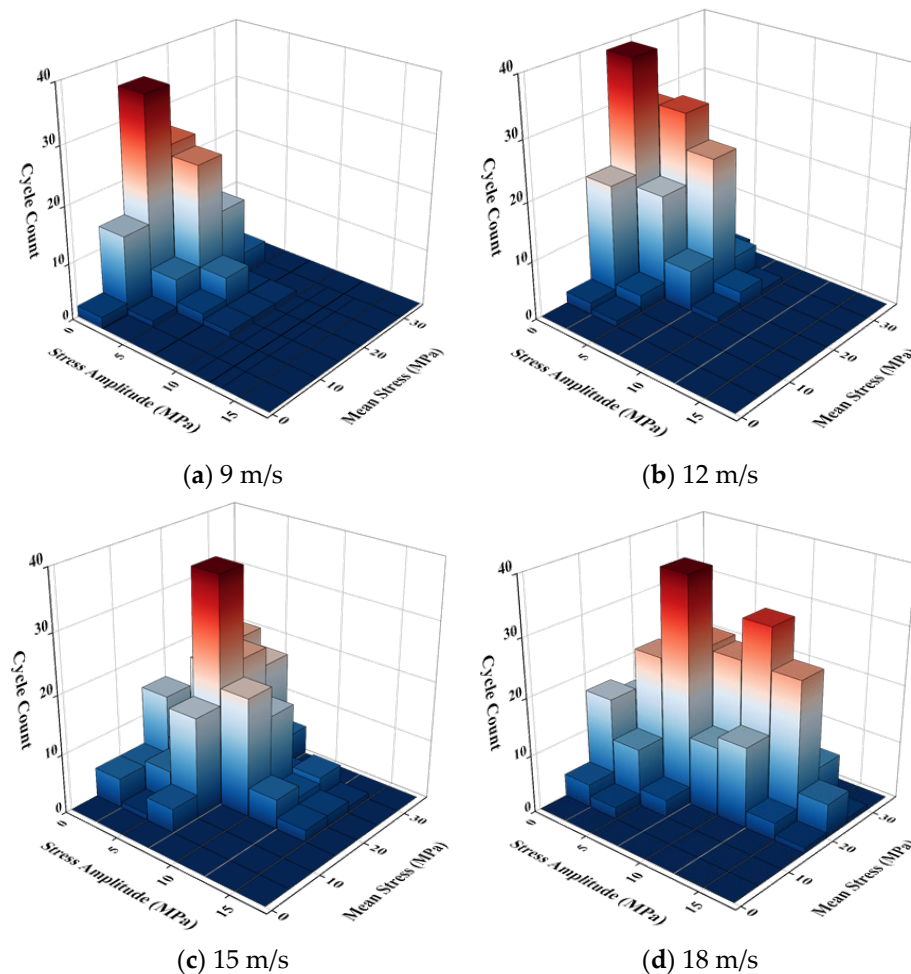


Figure 11. Load spectrum at observation region OR6.

Table 8. Fatigue life and annual damage of concrete at observation region OR6.

Wind Speed (m/s)	Applied Cycles	Cycles to Failure	Annual Damage Contribution
3.0	7.96×10^6	6.86×10^{17}	1.16×10^{-11}
6.0	2.7×10^8	4.61×10^{16}	5.86×10^{-9}
9.0	3.11×10^8	1.97×10^{15}	1.58×10^{-7}
12.0	1.99×10^9	1.97×10^{12}	1.01×10^{-3}
15.0	1.19×10^9	7.37×10^{10}	1.61×10^{-2}
18.0	6.47×10^8	2.73×10^{10}	2.37×10^{-2}
Total annual damage: 4.09×10^{-2}		Fatigue life: 24.46 years	

Figure 12 presents the fatigue lives for all observation regions under the baseline case. Among these, OR6 exhibits the shortest fatigue life, followed by OR5, while OR1 and OR2 show much longer lives. This distribution indicates a clear localization of concrete fatigue damage along the tower height. Therefore, OR5 and OR6 are identified as the critical concrete regions under the baseline condition and are used as the main reference for the following prestress relaxation analysis.

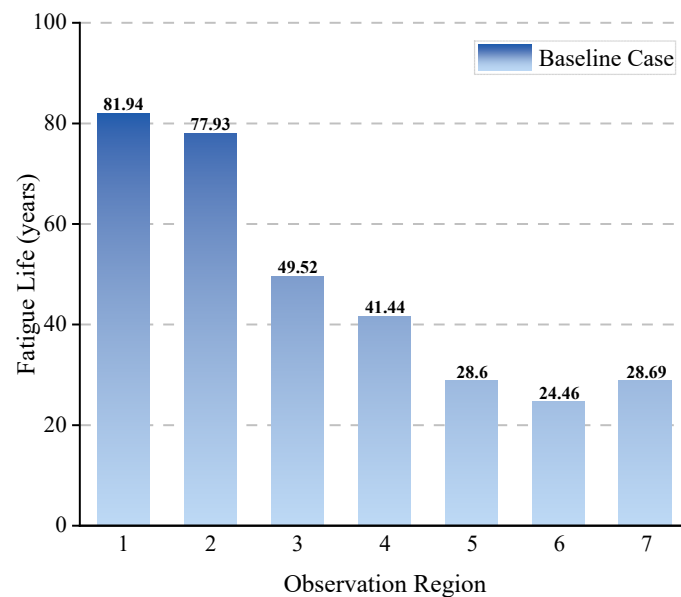


Figure 12. Fatigue life of concrete at observation regions under the baseline case.

5.2. Concrete Fatigue Response Under Tendon Relaxation

Figure 13 presents the fatigue-life distributions of the seven concrete observation regions under different relaxation cases. The relative ranking of the observation regions remains largely unchanged. In the single-tendon cases, OR5 and OR6 remain the regions with relatively short fatigue lives, especially in SR1 and SR2 (Figure 13a,b). OR6 exhibits the minimum fatigue life under almost all relaxation conditions, while OR1 and OR2 generally retain much longer lives. This indicates that the critical concrete region is relatively stable and is mainly controlled by its structural location.

Prestress relaxation therefore mainly amplifies the existing spatial difference in concrete fatigue performance. This feature becomes more evident in the three-tendon cases. As shown in Figure 13f,g, the fatigue lives of OR5 and OR6 decrease to only a few years, while clear reductions also occur in other observation regions. Similar behavior can be observed in Figure 13h,i. Thus, simultaneous multi-tendon relaxation extends the fatigue influence from the original critical regions to a larger portion of the concrete tower.

Some observation regions show small non-monotonic changes at intermediate prestress loss levels, such as those in Figure 13b–e. However, these local variations do not change the overall spatial pattern, and OR5 and OR6 remain the governing regions for concrete fatigue assessment.

5.3. Sensitivity Analysis of Concrete Fatigue Life

The sensitivity of concrete fatigue life is further examined with respect to the Weibull wind parameters and prestress relaxation. OR6 is selected as the response region because it governs the minimum concrete fatigue life under most conditions.

Figure 14 shows that the fatigue life of OR6 increases with the Weibull shape parameter k and decreases with the scale parameter λ . Within the investigated range, the predicted life varies from approximately 10 to more than 100 years. Concrete fatigue models generally consider the maximum and minimum stress levels rather than only the stress amplitude, and relatively small changes in the cyclic stress state can therefore lead to marked variations in fatigue life [32].

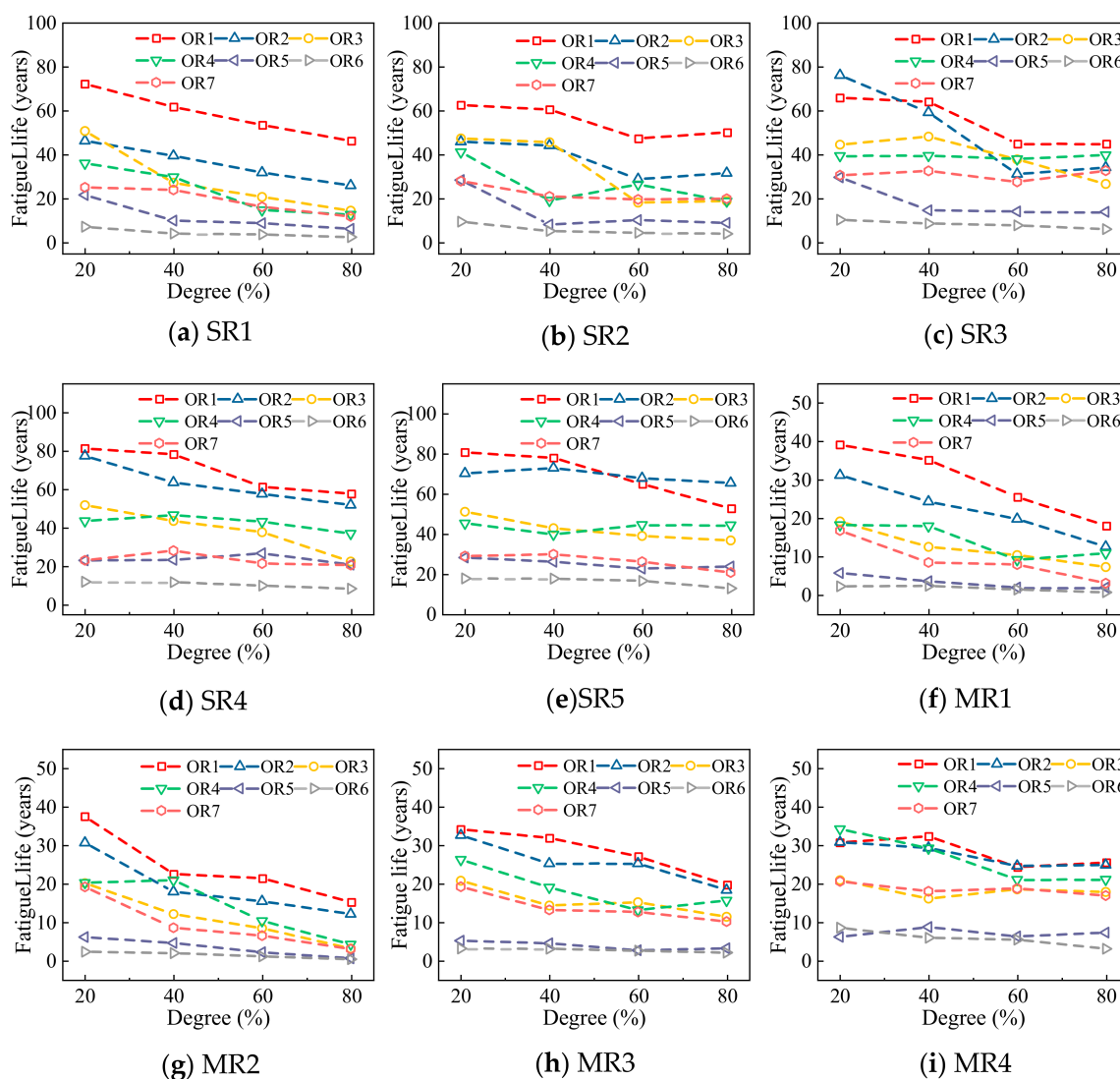


Figure 13. Fatigue life of concrete under various cases.

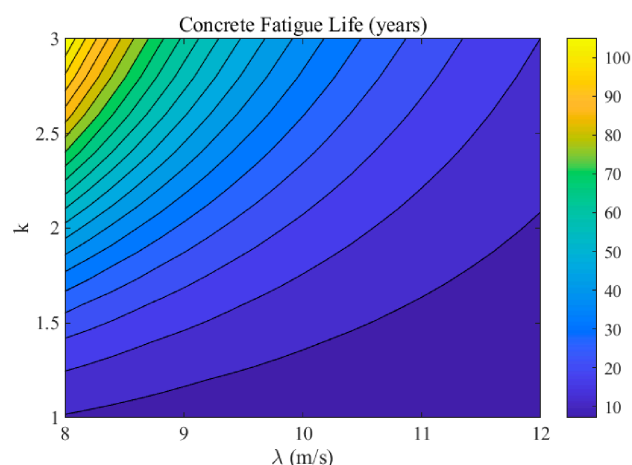


Figure 14. Effect of Weibull parameters on OR6 fatigue life under the baseline case.

Figure 15 shows that prestress relaxation has an even stronger influence on the available fatigue margin. The baseline minimum life of OR6 is 24.46 years, only slightly above the 20-year design service life. All relaxation cases reduce the minimum life below this reference level. At 80% relaxation, the minimum lives of SR1–SR5 are 2.29, 3.83, 5.94, 8.13,

and 12.77 years, respectively, while those of MR1–MR4 decrease further to 0.64, 0.31, 2.03, and 3.04 years. In particular, the minimum life in MR2 is only about 1.3% of the baseline value. These results show a nonlinear relationship between prestress loss and concrete fatigue life. As the effective prestress decreases, the cyclic stress state of the concrete changes and the available pre-compression is reduced. Higher effective prestress can improve fatigue resistance by limiting crack opening [42]. Therefore, prestress relaxation reduces this pre-compression reserve, leading to a rapid decrease in fatigue life once the remaining prestress becomes insufficient.

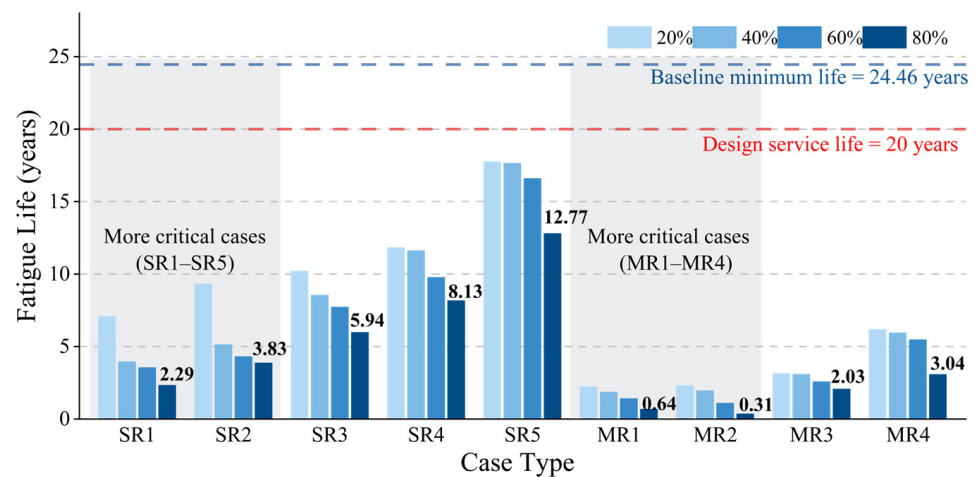


Figure 15. Minimum fatigue life of concrete observation regions.

6. Coupled Fatigue Life Analysis of the Overall Structure

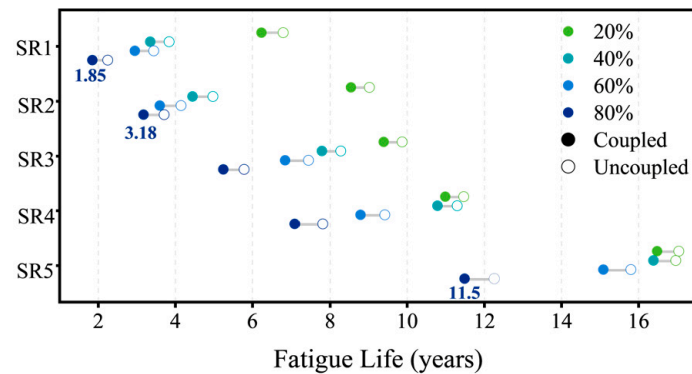
The bidirectionally coupled damage model developed in Section 2.4 is used to evaluate the overall fatigue life under different tendon relaxation cases. According to Sections 4.1 and 5.1, the baseline fatigue life of the critical concrete region OR6 is $T_c = 24.46$ years, and the minimum tendon fatigue life is $T_s = 39.90$ years. When the bidirectional damage coupling is neglected, $T = \frac{1}{\frac{\gamma_c}{T_c} + \frac{\gamma_s}{T_s}} = 23.05$ years. When bidirectional damage coupling is considered, the material stiffness is updated according to the damage state. Through finite element iterative analysis, the updated structural response and fatigue-relevant stress responses are obtained until the overall damage index satisfies $\gamma_c D_c + \gamma_s D_s \geq 1$. The corresponding cumulative service time of 22.15 years is then defined as the overall fatigue life under the coupled baseline condition.

6.1. Coupling Effect on Overall Fatigue Life

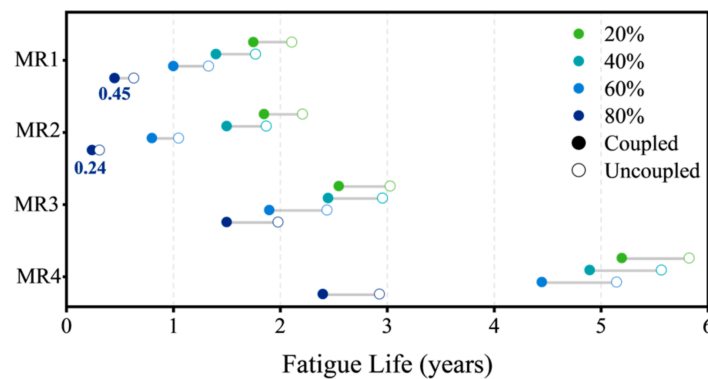
Figure 16 presents a comparison between the uncoupled and coupled overall fatigue lives under single-tendon and multi-tendon relaxation scenarios.

For the single-tendon relaxation cases (Figure 16a), the overall fatigue life is strongly affected by the relaxation position and prestress loss level. As prestress loss increases, the overall fatigue life decreases further. This trend is consistent with previous studies identifying the initial prestress level as an important factor in the fatigue assessment of post-tensioned concrete wind turbine towers [43]. SR1 and SR2, located closer to the windward side, exhibit more significant life reductions, whereas SR5 maintains a relatively longer fatigue life. At 80% prestress loss, the coupled fatigue lives of SR1 and SR2 decrease to 1.85 and 3.18 years, respectively, while SR5 retains a life of 11.50 years. This spatial dependence indicates that the effect of prestress loss on fatigue life is closely related to its location, consistent with previous observations of spatially different fatigue-critical regions in wind turbine towers [44].

The effect becomes more pronounced under multi-tendon relaxation (Figure 16b). At 80% prestress loss, the coupled fatigue lives of MR1 and MR2 decrease to 0.45 and 0.24 years, respectively. The stronger degradation associated with concentrated multi-tendon relaxation indicates that the spatial extent of prestress loss is another important factor controlling overall fatigue performance.



(a) Single-tendon relaxation cases



(b) Multi-tendon relaxation cases

Figure 16. Comparison of uncoupled and coupled overall fatigue lives under different tendon relaxation cases.

The design service life of the investigated tower is 20 years. Under critical relaxation cases, even a 20% prestress loss reduces the predicted overall fatigue life below this design target. Under severe multi-tendon relaxation, the coupled fatigue life further decreases to less than one year. These results indicate that relatively moderate prestress degradation may already have a significant influence on long-term fatigue performance, while concentrated high-level loss causes a marked reduction in the remaining fatigue life.

6.2. Variation in Coupling Life Reduction Rate

To further distinguish the influence of prestress loss itself from that of bidirectional damage coupling between materials on the overall fatigue life, Figure 17 presents the fatigue life reduction rate caused by bidirectional coupling. The baseline case gives a coupling-induced fatigue-life reduction of 3.9%, which provides a reference for evaluating the additional life reduction induced by coupling under different relaxation conditions.

For the single-tendon relaxation cases, the coupling life reduction rate generally increases with the prestress loss ratio. SR1 shows the strongest increase, while SR3–SR5 exhibit relatively smaller changes. This result indicates that the relaxation position affects not only the fatigue life but also the intensity of bidirectionally coupled damage.

The coupling effect is generally stronger under multi-tendon relaxation. The reduction rates of the MR cases are higher than those of most single-tendon cases over the investigated loss range, with MR1 showing the largest reduction of 28.6% at 80% prestress loss. Even MR4, which exhibits the smallest reduction within the multi-tendon group, reaches levels comparable to the more critical single-tendon cases. This suggests that simultaneous degradation of multiple tendons increases the contribution of damage interaction to the overall fatigue-life reduction.

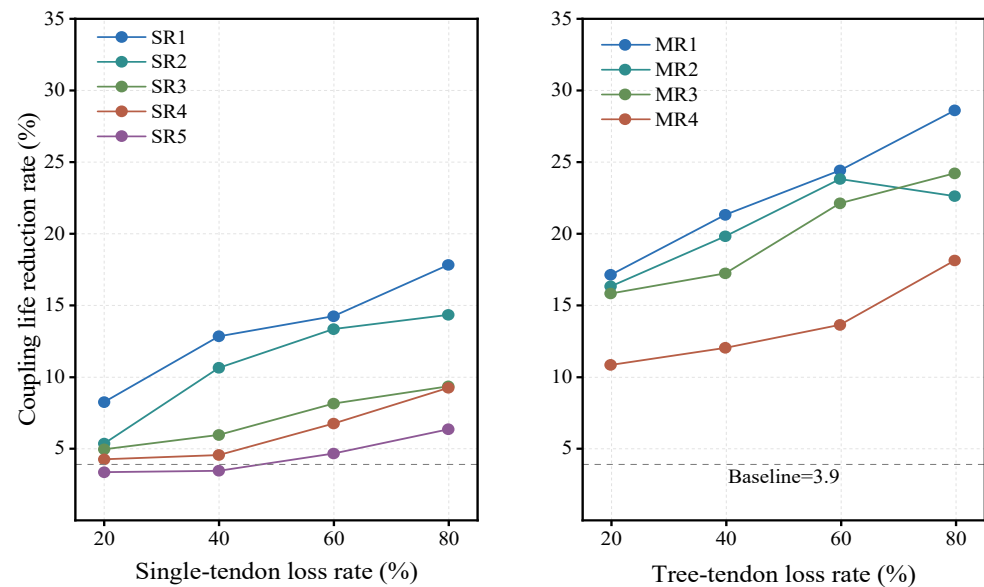


Figure 17. Variation in coupling reduction amplitude with prestress loss.

The increase in coupling reduction rate is not strictly monotonic for every scenario. For example, MR2 increases from 20% to 60% prestress loss but shows a slight decrease at 80%. This local variation may result from differences in the evolution rates of tendon and concrete damage during the coupling process. Therefore, the coupling effect is controlled by both the prestress loss level and the relative evolution of tendon and concrete damage, rather than by the loss percentage alone.

7. Deep Learning-Based Fatigue Life Assessment

7.1. The Proposed STPI-Net Framework

To address the high computational cost of conventionally coupled damage simulations and rapidly quantify how local component deterioration cascades to induce global structural failure, a Spatial-Topology and Physics-Informed Cascade Network (STPI-Net) is proposed for multi-scale fatigue life assessment of steel-prestressed concrete hybrid towers.

As shown in Figure 18, the established framework operates under a dual-space collaborative prediction paradigm. Specifically, component-level degradation prediction is implemented in a smoothed logarithmic fatigue space to stabilize gradient descent optimization during multi-task joint training. Subsequently, a nonlinear inverse transformation layer maps the intermediate logarithmic prediction results back to the linear physical time domain.

MULTI-TASK STRUCTURAL DEGRADATION AND STABILITY PREDICTION SYSTEM

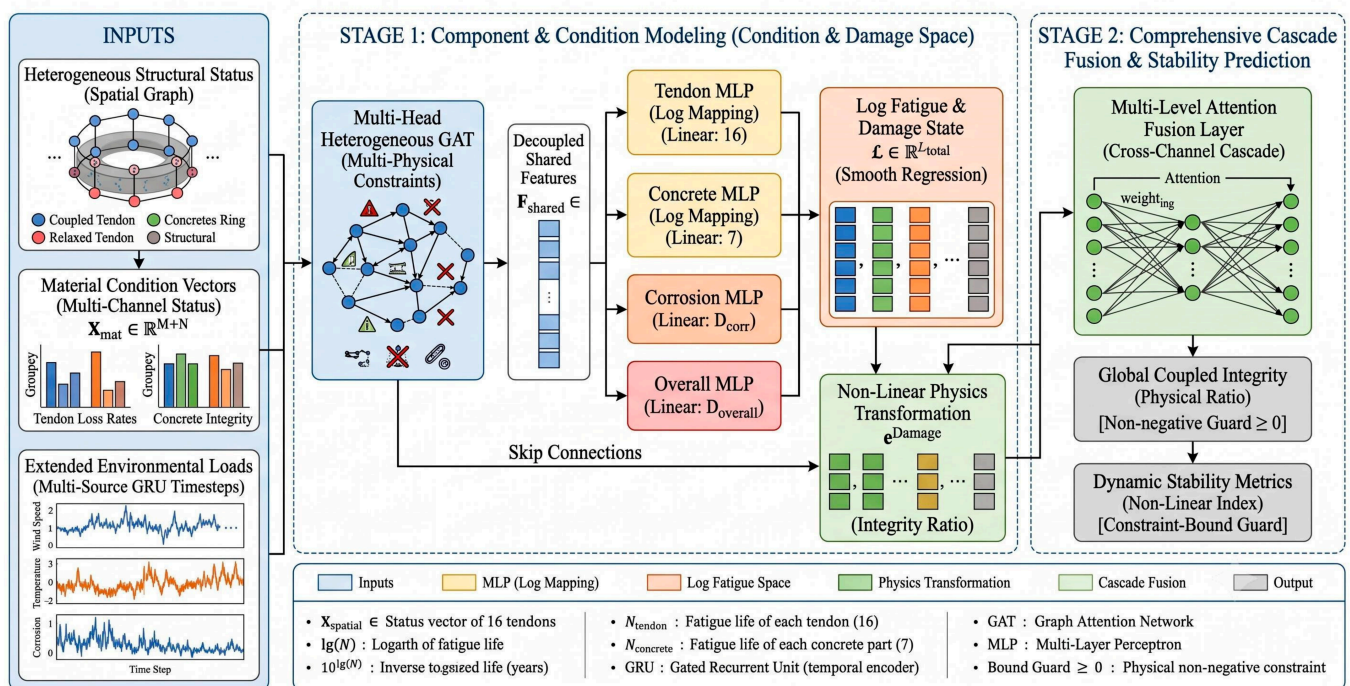


Figure 18. Configuration of the proposed STPI-Net.

(1) Spatial Topology Input Vectorization

For the anchoring zone containing 16 prestressed tendons, the health state of each tendon is determined by its loss rate, defined as $\alpha_i \in [0, 1]$. Therefore, STPI-Net converts the health state of 16 prestressing tendons into a 16-dimensional spatial input vector:

$$\mathbf{X}_s = [x_1, x_2, \dots, x_{16}]^T \in \mathbb{R}^{16} \quad (41)$$

where $x_i = 1 - \alpha_i$, $\alpha_i \in [0, 1]$ is the prestress loss ratio of tendon i , and x_i denotes the residual effective bearing capacity. For instance, $x_i = 1.0$ corresponds to the full working state of the tendon.

Then, $\mathbf{X}_s$ is fed into three shared ReLU-activated embedding layers to extract a 32D unified topology feature F :

$$F = \sigma(W_3 \cdot \sigma(W_2 \cdot \sigma(W_1 \cdot \mathbf{X}_s + b_1) + b_2) + b_3) \quad (42)$$

In this formula, $W_1 \in \mathbb{R}^{64 \times 16}$, $W_2 \in \mathbb{R}^{64 \times 64}$, $W_3 \in \mathbb{R}^{32 \times 64}$ are layer weights; b_1, b_2, b_3 are bias vectors; $\sigma(\cdot)$ = ReLU activation. This step filters geometric noise and outputs shared features for multi-task prediction.

(2) Component Degradation Modeling (Stage 1)

Wide-ranging fatigue life values trigger severe gradient oscillation during direct regression. To stabilize training, component fatigue prediction is implemented in logarithmic fatigue space with two parallel MLP branches sharing the feature F . For the prestressed tendon prediction branch, the prediction formula is defined as

$$\hat{Y}_{tendon}^{log} = W_{tendon} \cdot F + b_{tendon} \in \mathbb{R}^{16}, \hat{Y}_{tendon,i}^{log} = lg \hat{N}_{tendon,i} \quad (43)$$

where W_{tendon} and b_{tendon} are the branch-specific weights and bias; $\hat{N}_{tendon,i}$ denotes the numerical value of the predicted fatigue life of i -th tendon in years.

Correspondingly, the concrete logarithmic life branch (7 observed regions) is expressed as

$$\hat{Y}_{\text{concrete}}^{\log} = W_{\text{concrete}} \cdot F + b_{\text{concrete}} \in \mathbb{R}^7, \hat{Y}_{\text{concrete},j}^{\log} = \lg \hat{N}_{\text{concrete},j} \quad (44)$$

Logarithmic mapping compresses the value span of fatigue life, smooths optimization gradients, and improves model generalization for diverse damage conditions.

(3) Physics-Informed Inverse Transformation Layer

Logarithmic outputs serve only for training optimization and cannot be directly used for engineering evaluation. An exponential inverse transformation layer recovers logarithmic predictions to real service life in years:

$$\hat{Y}_{\text{tendon}}^{\text{years}} = 10^{\hat{Y}_{\text{tendon}}^{\log}} \in \mathbb{R}^{16} \quad (45)$$

$$\hat{Y}_{\text{concrete}}^{\text{years}} = 10^{\hat{Y}_{\text{concrete}}^{\log}} \in \mathbb{R}^7 \quad (46)$$

This step aligns network outputs with physical time scales and supplies component residual life data for subsequent global coupling prediction.

(4) Global Cascade Coupling Prediction (Stage 2)

This stage predicts overall tower coupled fatigue life by capturing mutual degradation interactions between tendons and concrete. The original spatial vector and converted physical life results are concatenated to construct an augmented feature matrix:

$$X_{ca} = \begin{bmatrix} X_s & | & \hat{Y}_{\text{tendon}}^{\text{years}} & | & \hat{Y}_{\text{concrete}}^{\text{years}} \end{bmatrix} \in \mathbb{R}^{39} \quad (47)$$

A two-layer fully connected fusion network integrates cross-channel features to generate preliminary global coupled life:

$$y_{co} = W_{c2} \cdot \sigma(W_{c1} \cdot X_{ca} + b_{c1}) + b_{c2} \quad (48)$$

where W_{c1}, W_{c2} and b_{c1}, b_{c2} are the fusion network's parameters. To enforce physical plausibility, a physical boundary constraint eliminates unphysical negative-life predictions via lower-bound truncation:

$$\hat{Y}_{co} = \max(0.05, y_{co}) \quad (49)$$

The threshold of 0.05 years guarantees that all outputs comply with structural mechanical laws and engineering reality.

The model training is optimized by the Adam optimizer, with a fixed learning rate of 0.003 and a total training epoch of 500. A unified multi-task objective loss function is constructed to synchronously optimize the three regression tasks of tendon logarithmic life prediction, concrete logarithmic life prediction, and system coupled life prediction. The total loss is defined as the sum of the mean squared errors (MSEs) of all subtasks:

$$\mathcal{L}_{\text{total}} = \text{MSE}(\hat{Y}_{\text{tendon}}^{\log}, Y_{\text{tendon}}^{\log}) + \text{MSE}(\hat{Y}_{\text{concrete}}^{\log}, Y_{\text{concrete}}^{\log}) + \text{MSE}(\hat{Y}_{co}^{\text{norm}}, Y_{co}^{\text{norm}}) \quad (50)$$

where the superscript "norm" indicates that the coupled target values are standardized.

7.2. Experiments and Validations

All training and validation samples adopted in this study are generated by high-precision finite element method (FEM) numerical simulations. The dataset covers typical engineering working conditions, which are intended to represent operational damage characteristics of structural systems.

To improve the dataset's coverage of different prestress loss patterns, this study further expands the prestress relaxation scenarios. For the anchorage region containing 16 prestressing strands, single-strand loss conditions are first considered, comprising 16 spatial position combinations in total. Subsequently, cases involving the simultaneous loss of two or three strands are further considered. Adjacent strand combinations, opposite-position strand combinations, and other random combinations are included. These scenarios simulate local prestress degradation under different spatial distribution forms. They also cover the structural response characteristics associated with the simultaneous degradation of multiple strands.

Considering different strand loss forms, a total of 69 different prestress loss topology combinations are generated. Combined with four prestress loss levels of 20%, 40%, 60%, and 80%, 276 independent damage scenarios are ultimately formed for the training and validation of the STPI-Net model. This expanded dataset can cover structural degradation patterns under different spatial topology forms and loss degrees, enabling the model to learn the nonlinear relationship between strand spatial distribution, prestress loss level, and fatigue life degradation.

This study adopts a scenario-based five-fold cross-validation strategy. Specifically, working conditions with the same spatial loss topology characteristics are grouped into the same data group and kept intact during data partitioning. In each fold of validation, the validation set contains complete damage scenarios that are not involved in model training. Therefore, model evaluation not only reflects its fitting ability to known working conditions but also examines its prediction ability for unknown prestress loss combinations.

For each cross-validation fold, the mean μ_{train} and standard deviation σ_{train} are calculated using only its training subset. The same values are then used to standardize both the training and held-out validation subsets according to Equation (51):

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}} \quad (51)$$

In terms of the validation, the coefficient of determination (R^2) and mean absolute error (MAE) are selected as quantitative evaluation indices to comprehensively assess the out-of-sample prediction performance of STPI-Net across all cross-validation folds.

7.3. Numerical Results and Discussion

Figure 19 presents the logarithmic fatigue life prediction performance for 16 prestressing tendons and 7 critical concrete observation regions in Stage 1. For tendon prediction, nearly all output channels achieve out-of-sample $R^2 > 0.8$, with the overall metric $R^2 = 0.8879$ and $\text{MAE} = 0.1409$. The scattered sample points closely align with the ideal $y = x$ prediction line, which verifies that the shared spatial topology embedding module (Equation (42)) effectively eliminates geometric interference induced by differential prestress relaxation positions and extracts the intrinsic degradation law of each tendon.

For concrete components covering the base ring A0 to high-position Part 30 segments, the model yields a higher prediction accuracy of $R^2 = 0.9030$ and $\text{MAE} = 0.0764$. The significant improvement in fitting effect demonstrates that logarithmic-domain regression (Equations (43) and (44)) narrows the wide value range of fatigue life, stabilizes gradient backpropagation during multi-task training, and accurately characterizes the highly nonlinear fatigue deterioration of concrete under coupled wind loads and prestress loss. Minor deviations of individual sample points are mainly attributed to extreme prestress loss combinations that trigger abrupt tensile stress surges in concrete, which introduce strong nonlinear disturbances to structural stress responses.

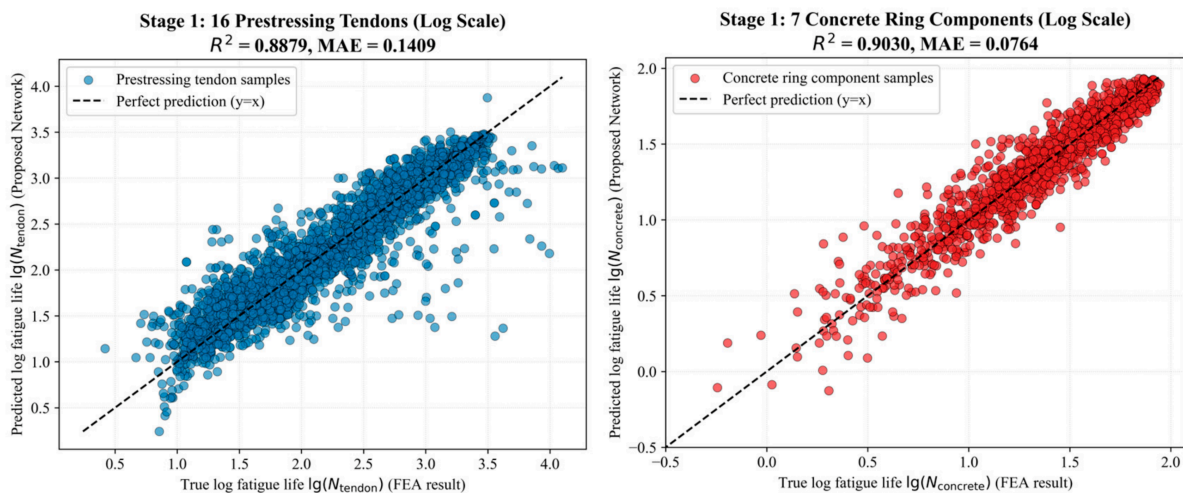


Figure 19. Component prediction performance.

Stage 2 realizes the prediction of overall tower coupled fatigue life by fusing original spatial input and converted physical life outputs of all components (Equation (47)). As illustrated in Figure 20, the cross-validation results for system-level prediction deliver an overall $R^2 = 0.8195$ and MAE = 1.9009. Most FEA simulation samples are distributed along the perfect prediction line, proving that the concatenated augmented feature X_{ca} can capture the bidirectional mutual degradation coupling between tendons and concrete.

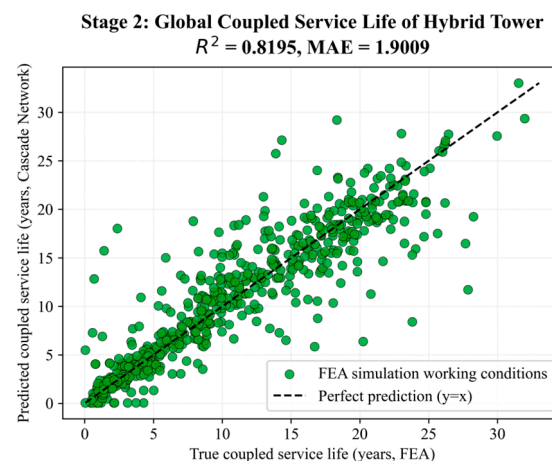


Figure 20. System prediction performance.

Compared with component-level prediction, the global coupled life exhibits slightly lower accuracy, which is reasonable for two core reasons. First, the overall structural failure criterion is governed by the coupling of two materials, introducing stronger nonlinear interaction effects that increase prediction difficulty. Second, the random wind speed Weibull distribution amplifies the stochastic fluctuation of long-term cumulative fatigue damage, bringing unavoidable uncertainty to the ground-truth labels generated by FEM iterations. Even so, the R^2 value close to 0.8 suggests that STPI-Net can provide reasonably acceptable global life predictions for the investigated cases, although further validation is needed before it can be regarded as reliable for engineering fatigue evaluation.

8. Engineering Implications and Limitations

8.1. Engineering Implications

The proposed fatigue assessment framework provides a basis for fatigue assessment and condition monitoring of steel–prestressed concrete hybrid wind turbine towers. The results indicate that windward-side tendons, adjacent tendon groups, and the middle and lower concrete regions are more sensitive to prestress relaxation. These regions may therefore be prioritized in tendon monitoring, fatigue inspection, and maintenance planning. For the investigated tower, 20% prestress loss can be used as a preliminary warning level because investigated cases with prestress loss reaching 20% result in fatigue life shorter than the 20-year design life. The spatial distribution and magnitude of prestress loss obtained from monitoring data can be used as inputs to STPI-Net for estimating component and overall fatigue life, thereby linking prestress-loss identification with structural fatigue assessment.

8.2. Limitations and Future Research

The fatigue life of a hybrid wind turbine tower is affected by its material properties, structural configuration, and site-specific wind loading. Therefore, the results of this study should be interpreted mainly as fatigue-life trends under the investigated prestress-loss cases. Application to other tower types requires appropriate parameter updating and verification.

The study mainly considers tendon relaxation as the source of prestress loss. In practice, prestress loss may also result from concrete shrinkage and creep, among other long-term mechanisms. The 60% and 80% prestress-loss cases are introduced as severe parametric scenarios to examine sensitivity and identify critical degradation states. They are useful for evaluating the progressive reduction in fatigue performance under high loss levels, although their quantitative results should be interpreted within the assumptions of the adopted structural model. Future studies should consider combined prestress-loss mechanisms and further examine the structural response under severe degradation.

The STPI-Net training and validation data are mainly generated from finite element simulations. Experimental and field-monitoring data can be introduced to calibrate the model, account for structural and loading variability, and further improve its generalization and reliability.

9. Conclusions

This study established a numerical and analytical framework to evaluate the long-term fatigue performance of a steel–prestressed concrete hybrid wind turbine tower under different tendon relaxation cases. By integrating a novel stiffness-feedback bidirectional damage model with a deep learning surrogate, the fatigue performance of the structure was investigated. The main conclusions are as follows:

- (1) Prestress relaxation reduces the fatigue performance of both prestressing tendons and concrete. At 80% prestress loss, the minimum tendon fatigue life decreases to 11.48 years under windward-side single-tendon relaxation and to 4.66 years under concentrated multi-tendon relaxation. OR5 and OR6 remain the critical concrete regions, with the minimum concrete fatigue life decreasing to 0.31 years under severe multi-tendon relaxation. These results show that fatigue performance depends not only on the level of prestress loss, but also on its location and spatial distribution. These findings highlight the need to prioritize windward-side regions and adjacent tendon groups in fatigue monitoring and inspection.
- (2) The bidirectionally coupled damage model captures a progressive feedback effect that further reduces the overall fatigue life of the tower. Concrete stiffness degrada-

tion increases the stress demand in the tendons, while tendon prestress loss reduces concrete pre-compression and accelerates concrete damage. As a result, the coupled fatigue life falls below the 20-year design life even at 20% prestress loss. In the MR1 case at 80% prestress loss, it decreases to 0.45 years, which is 28.6% lower than the corresponding uncoupled fatigue life. These findings show that damage coupling causes an additional reduction in overall fatigue life, and this effect becomes more significant under concentrated multiple-tendon relaxation.

- (3) The proposed Spatial-Topology and Physics-Informed Cascade Network (STPI-Net) incorporates spatial tendon configurations and relaxation conditions. Scenario-based five-fold cross-validation gives $R^2 = 0.8879$ for logarithmic tendon fatigue life, $R^2 = 0.9030$ for logarithmic concrete fatigue life, and $R^2 = 0.8195$ for overall coupled fatigue life. The results demonstrate the potential of the proposed model for fatigue assessment based on prestress loss data.

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